

# Machine Learning-Based Post-Hoc Calibration of Low-Cost Air Quality Sensors Using Reference Monitoring Stations

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| <b>Cycle</b>               | : Second Cycle                                                               |
| <b>Course name</b>         | : Thesis for Master Degree in Micro Data Analysis                            |
| <b>Course Code</b>         | : MI4001                                                                     |
| <b>Credits</b>             | : 30 Credits                                                                 |
| <b>Supervisors</b>         | : Professor Yves Rybarczyk, Daniel de Amaral da Silva                        |
| <b>School</b>              | : School of Information and Engineering                                      |
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| <b>Date of Examination</b> | : 2026-June-04                                                               |

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# Abstract

Air pollution is a major environmental and public health challenge worldwide. Fine particulate matter (PM<sub>2.5</sub>) is particularly harmful because it can penetrate deep into the lungs and bloodstream, contributing to respiratory and cardiovascular diseases. Although regulatory-grade reference monitoring stations provide accurate measurements, their high-cost limits spatial coverage. Low-cost sensors (LCS) offer a cost-effective alternative but require calibration to improve measurement reliability.

This thesis presents a machine learning-based post-hoc calibration framework for low-cost PM<sub>2.5</sub> sensors using reference monitoring station data. A Linear Tree Regressor was selected because it combines nonlinear decision-tree partitioning with interpretable local linear regression models. The study used the SensEURCity dataset collected in Antwerp, Belgium, comprising approximately 18.7 million minute-level observations from 34 low-cost sensors and 9 reference monitoring stations. PM<sub>2.5</sub> concentration, relative humidity, temperature, and atmospheric pressure were used as model inputs.

Three research questions were investigated: (1) whether machine learning improves calibration accuracy compared with the U.S. Environmental Protection Agency (EPA) correction method, (2) how many reference stations are required for effective calibration, and (3) how spatial distance affects calibration error.

The Linear Tree Regressor achieved  $MAE = 2.31$ ,  $RMSE = 3.91$ , and  $R^2 = 0.83$ , outperforming the EPA correction method ( $MAE = 3.55$ ,  $RMSE = 5.29$ ,  $R^2 = 0.69$ ). The EPA performance was consistent with previous validation studies, while the proposed model further improved calibration accuracy. Results also showed that effective calibration could be achieved using a limited number of reference stations. Furthermore, the proposed model maintained lower prediction error than the EPA method across most practical urban deployment distances. These findings demonstrate the potential of machine learning to support accurate, scalable, and cost-effective urban air quality monitoring.

## Keywords

PM<sub>2.5</sub>, Low-Cost Sensors, Environmental Monitoring, Sensor Calibration, Air Pollution, Linear Tree Regressor, Machine Learning, EPA Correction Method, Urban Air Quality, Reference Monitoring Stations, ANCOVA

# Acknowledgement

We would like to thank our supervisors, Professor Yves Rybarczyk and PhD researcher Daniel de Amaral da Silva, for their constant guidance, valuable feedback and support during this research project. The success of this thesis is largely due to their expertise in artificial intelligence, environmental monitoring, data science, and machine learning.

We are also sincerely grateful to our examiner, Moudud Alam, for his valuable guidance and constructive feedback throughout the course, particularly from the early stages of the research to the midterm evaluation, which significantly contributed to the development and quality of this thesis.

We would also like to thank Dalarna University for the opportunity and resources to carry out this research as part of the Master's Programme in Data Science.

We would also like to express our gratitude to all collaborators and researchers who contributed to the development of the air quality dataset used in this study.

Finally, we sincerely thank our family and friends for their continuous encouragement, patience, and support throughout our studies and the completion of this thesis.

# List of Abbreviations

|                                            |                                           |
|--------------------------------------------|-------------------------------------------|
| <b>ANCOVA</b>                              | Analysis of Covariance                    |
| <b>CSV</b>                                 | Comma-Separated Values                    |
| <b>dbt</b>                                 | Data Build Tool                           |
| <b>EPA</b>                                 | U.S. Environmental Protection Agency      |
| <b>JRC</b>                                 | Joint Research Centre                     |
| <b>LCS</b>                                 | Low-Cost Sensor                           |
| <b>MAE</b>                                 | Mean Absolute Error                       |
| <b>OPCN3</b>                               | Optical Particle Counter N3               |
| <b>PM</b>                                  | Particulate Matter                        |
| <b>PM<sub>1</sub></b>                      | Particulate Matter $\leq 1 \mu\text{m}$   |
| <b>PM<sub>2.5</sub></b>                    | Particulate Matter $\leq 2.5 \mu\text{m}$ |
| <b>PM<sub>10</sub></b>                     | Particulate Matter $\leq 10 \mu\text{m}$  |
| <b>RH</b>                                  | Relative Humidity                         |
| <b>RMSE</b>                                | Root Mean Squared Error                   |
| <b>R<sup>2</sup></b>                       | Coefficient of Determination              |
| <b>RQ</b>                                  | Research Question                         |
| <b>SD</b>                                  | Standard Deviation                        |
| <b>SQL</b>                                 | Structured Query Language                 |
| <b><math>\mu\text{g}/\text{m}^3</math></b> | Micrograms per Cubic Meter                |

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# 1. Introduction

## 1.1 Background

Air pollution has emerged as one of the most significant environmental and public health challenges affecting modern urban environments. Rapid industrialization, population growth, urban traffic congestion, residential heating systems, and construction activities continuously contribute to increasing levels of atmospheric pollution [3],[17],[23]. The negative impact of air pollution on human health has been extensively documented in environmental and medical research. Exposure to polluted air is associated with respiratory diseases, cardiovascular complications, reduced lung function, asthma, chronic obstructive pulmonary disease (COPD), and premature mortality [3],[5]. Exposure to polluted air is associated with millions of premature deaths worldwide each year [3].

Among air pollutants, particulate matter smaller than 2.5 micrometers ( $PM_{2.5}$ ) is considered particularly hazardous because its microscopic size allows it to penetrate deeply into the human respiratory system. In contrast to bigger particles,  $PM_{2.5}$  has the ability to bypass the natural defense system of the respiratory tract and reach the alveolar regions of the lung [17],[18],[19]. In extreme cases, these particles can even get into the bloodstream and interfere with the cardiovascular system. Long term exposure to  $PM_{2.5}$  has been associated with lung cancer, heart disease, stroke and decreased life expectancy [3]. Therefore, precise monitoring of  $PM_{2.5}$  concentrations is critical for environmental protection, urban planning, epidemiological studies and public health management.

The traditional air quality monitoring system is based on scientific-grade reference monitoring stations. These stations use highly accurate instruments such as TEOM analyzers, gravimetric samplers, beta attenuation monitors and optical particle counters to provide regulatory grade measurements. Reference stations are generally considered the gold standard for air quality monitoring, due to their accuracy and reliability. However, these systems are very expensive to install, calibrate and maintain, but they are accurate [5],[7]. The high infrastructure cost severely limits the number of reference stations cities can deploy. This leads to limited spatial coverage for monitoring in many urban areas, limiting the understanding of local pollution dynamics.

Low-cost air quality sensors have become a viable alternative for dense urban air quality monitoring in recent years. While regulatory-grade instruments such as the Tapered Element Oscillating Microbalance (TEOM), Beta Attenuation Monitor (BAM), GRIMM aerosol spectrometers, and Palas optical particle counters provide highly accurate measurements, their high installation and maintenance costs limit large-scale deployment. In contrast, low-cost sensors such as the Plantower PMS5003, which is widely used in AirSenseEUR and PurpleAir monitoring systems, offer an affordable and scalable solution for dense spatial monitoring but require calibration to improve measurement accuracy [11].

Due to their relatively low cost, compact size, and ease of deployment, hundreds of low-cost sensors can be distributed across urban environments to provide high-resolution spatial and temporal air quality information [3], [5], [10], [23], [24]. Such dense monitoring networks support a variety of applications, including smart-city initiatives, citizen science projects, exposure assessment, environmental research, and real-time pollution monitoring [3], [10].

Despite these advantages, low-cost sensors have important limitations that reduce their reliability. Measurements obtained from low-cost sensors are often affected by environmental conditions such as temperature, humidity, smoke and dust [3],[4],[6]. In addition, sensor drift, electronic noise, aging effects, and manufacturing variability can significantly reduce measurement quality over time. Two sensors of the same model may produce noticeably different readings under identical environmental conditions. Consequently, raw low-cost sensor measurements cannot generally be used directly for scientific or regulatory purposes without correction.

To address these limitations, calibration techniques are required. Calibration is the adjustment of low-cost sensor measurements to better match measurements from reference monitoring stations. Conventional calibration methods usually take linear corrections or empirical formulas, such as the correction method of U.S. Environmental Protection Agency (EPA). However, these methods may not well characterize the complex nonlinear relationships between sensor outputs and environmental variables.

Machine learning techniques provide a promising alternative because they can model nonlinear relationships and interactions between environmental variables more effectively than traditional correction methods [12],[13]. Thus, this thesis investigates the question as to whether machine learning-based post-hoc calibration methods can improve the reliability and usability of low-cost air quality sensor measurements in urban environments.

## 1.2 Problem Statement

Low-cost PM<sub>2.5</sub> sensors offer a scalable approach for urban air quality monitoring, but their measurements are frequently unreliable due to environmental influences and sensor limitations. Existing calibration approaches such as the EPA correction formula provide partial improvements but may fail to capture complex nonlinear relationships between sensor measurements and environmental variables.

Furthermore, urban deployments often contain only a limited number of reference monitoring stations because of infrastructure costs. It is therefore important to determine:

1. Whether machine learning methods can outperform traditional correction algorithms.
2. How many reference stations are truly necessary.
3. Whether the distance between low-cost sensors and reference monitoring stations affects calibration quality, how calibration error changes with increasing distance, and at what distance the effect becomes practically meaningful.

## 1.3 Research Aim

The primary aim of this thesis is to develop and evaluate a machine learning-based post-hoc calibration framework for low-cost air quality sensors using reference monitoring station measurements.

The Linear Tree Regressor was selected because it combines the flexibility of decision-tree partitioning with the interpretability of linear regression models. Unlike traditional black-box machine learning approaches, such as Random Forests, Gradient Boosting, and deep learning models, the Linear Tree Regressor produces explicit local linear equations within each leaf node,

making the calibration process easier to understand and interpret. This characteristic is particularly valuable in environmental monitoring applications, where transparency and explainability are important. Furthermore, the model provides a suitable balance between predictive performance and interpretability, allowing a meaningful comparison with the EPA correction approach. Although more advanced machine learning models have been applied successfully for air quality sensor calibration, evaluating such models is outside the scope of this thesis and is suggested as a direction for future research.

The study investigates not only the number of reference monitoring stations required for effective calibration but also the potential influence of spatial distance between low-cost sensors and reference stations. Distance is therefore treated as an important spatial factor and is examined separately in RQ3.

## 1.4 Research Questions

- RQ1 – Can a Linear Tree Regressor improve  $PM_{2.5}$  calibration accuracy compared to the EPA correction method?
- RQ2 – How many reference monitoring stations are needed to achieve reliable calibration performance?
- RQ3 – How does the distance between low-cost sensors and reference monitoring stations affect calibration quality, how does calibration error change with increasing distance, and at what distance does the effect become practically meaningful?

## 1.5 Research Objectives

Followings are the objectives.

- To implement the EPA  $PM_{2.5}$  correction method as a baseline calibration approach.
- To develop a machine learning-based Linear Tree Regressor calibration model.
- To evaluate calibration performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ( $R^2$ ). These metrics are calculated using stratified five-fold cross-validation, and the reported values represent the mean performance across the five validation folds.
- To investigate the effect of reference monitoring station subsets (different combinations and numbers of reference stations) on calibration performance.
- To analyze the effect of spatial distance on prediction error using regression analysis and ANCOVA.
- To evaluate future opportunities for cross-city model generalization.

## 2. Literature Review

### 2.1 Introduction

This section presents a review of previous studies related to air pollution monitoring, low-cost air quality sensors, calibration techniques for  $PM_{2.5}$ , machine learning-based calibration methods, and spatial air quality analysis. The review emphasizes studies that have explored the challenges in the accuracy of low-cost sensors and the application of statistical and machine learning approaches to improve particulate matter measurements.

### 2.2 Air Pollution and $PM_{2.5}$

Air pollution continues to create major environmental and health problems in many urban regions around the world. Air pollution contributes to millions of premature deaths each year and is strongly associated with respiratory and cardiovascular diseases worldwide [3].

Among the air pollutants, particulate matter with diameter less than 2.5 micrometers, also known as  $PM_{2.5}$ , is believed to be particularly harmful due to their very small particle size. Due to their microscopic size,  $PM_{2.5}$  particles can pass through normal respiratory filtering mechanisms and reach deep lung tissues [17],[18],[19].

The  $PM_{2.5}$  pollution has many natural and anthropogenic sources including vehicle emissions, industrial processes, biomass burning, domestic heating systems, construction activities and secondary atmospheric chemical reactions [3]. Urban areas are especially vulnerable due to the high density of traffic and concentration of industrial activities.

Previous studies have shown that long-term exposure to  $PM_{2.5}$  is significantly associated with serious health problems including cardiovascular disease, chronic obstructive pulmonary disease (COPD), asthma, respiratory infections, and lung cancer [3]. Therefore, accurate monitoring of  $PM_{2.5}$  concentrations is very important for environmental protection, public health management, and urban policy making [3].

### 2.3 Traditional Air Quality Monitoring Systems

Traditional air quality monitoring systems rely on regulatory-grade reference monitoring stations equipped with highly accurate scientific instruments. Common technologies used in these stations include beta attenuation monitors (BAM), tapered element oscillating microbalance (TEOM) analyzers, optical particle counters, and gravimetric measurement systems. These instruments are designed to provide precise and standardized measurements of atmospheric pollutants and are widely considered the gold standard for air quality monitoring [5],[7].

Reference monitoring stations provide highly reliable measurements but are expensive to install operate and maintain. The instruments require controlled operating environments, regular calibration procedures, specialized maintenance and continuous technical supervision. As a result, typically only a limited number of reference stations can be deployed within a city or region.

Traditional monitoring networks have limited spatial coverage, posing a major barrier to understanding localized pollution patterns in urban environments. Air pollution levels are highly

variable over short distances, owing to traffic density, industrial activity, meteorological conditions and urban morphology [8],[9]. Sparse reference station networks may not fully capture fine-scale spatial variability in PM<sub>2.5</sub> concentrations.

## 2.4 Low-Cost Air Quality Sensors

In the last few years low-cost air quality sensors have become popular for urban environmental monitoring as they are cheap, portable and easy to deploy. Unlike traditional reference monitoring stations, these sensors can be deployed in large numbers to provide high-resolution spatial monitoring of air pollution across cities and communities.

Several practical advantages of low-cost sensor networks include lower cost of deployment, better spatial coverage, real-time monitoring, and flexible installation in a variety of environments. These features make low-cost sensors particularly attractive for smart-city applications, community-based monitoring efforts, and large-scale environmental sensing systems [3],[5],[10].

Despite these advantages, low-cost particulate matter sensors have several important limitations that affect measurement reliability. Sensor performance can be strongly influenced by environmental conditions such as relative humidity, temperature, smoke and dust [4],[6]. In addition, factors such as sensor aging, particle composition, atmospheric conditions, and environmental interference can introduce noise, drift and systematic bias in measurements.

Among the various low-cost particulate matter sensors, the Plantower PMS5003 particle sensor, manufactured by Plantower Co., Ltd., has become one of the most widely used sensors in urban PM<sub>2.5</sub> monitoring studies [11],[21]. The PMS5003 is an optical particle sensor commonly used in commercial monitoring platforms such as PurpleAir and has been extensively evaluated in previous calibration and validation studies [11],[21]. The sensor is popular because of its low cost, compact size, and reasonable performance after calibration [11],[20],[21]. Consequently, many studies have explored the calibration methods and machine learning techniques to improve the PMS5003 measurement accuracy under different environmental conditions.

Table 1. Comparison of Reference Monitoring Stations and Low-Cost Air Quality Sensors

| Feature                    | Reference Monitoring Stations | Low-Cost Sensors |
|----------------------------|-------------------------------|------------------|
| Measurement Accuracy       | Very high                     | Moderate to low  |
| Cost                       | Very expensive                | Low cost         |
| Maintenance                | High                          | Low to moderate  |
| Spatial Coverage           | Limited                       | High             |
| Calibration Requirement    | Minimal                       | High             |
| Portability                | Low                           | High             |
| Infrastructure Requirement | Complex                       | Simple           |
| Real-Time Monitoring       | Yes                           | Yes              |
| Sensitivity to Environment | Low                           | High             |
| Scalability                | Limited                       | High             |

## 2.5 EPA PM<sub>2.5</sub> Correction Method

The U.S. Environmental Protection Agency (EPA) proposed a correction algorithm for the Plantower PMS5003 particle sensor to improve PM<sub>2.5</sub> measurement accuracy under different environmental conditions [11]. The correction algorithm adjusts raw PM<sub>2.5</sub> measurements using relative humidity and concentration-based piecewise equations.

### EPA Correction Rules

#### Rule 1

If:  $AG_{raw} < 30$

$$PM_{2.5} = 0.524AG_{raw} - 0.0862RH_{raw} + 5.75$$

#### Rule 2

If:  $30 \leq AG_{raw} < 50$

$$PM_{2.5} = [0.786(AG_{raw}/20 - 3/2) + 0.524(1 - (AG_{raw}/20 - 3/2))]AG_{raw} - 0.0862RH_{raw} + 5.75$$

#### Rule 3

If:  $50 \leq AG_{raw} < 210$

$$PM_{2.5} = 0.786AG_{raw} - 0.0862RH_{raw} + 5.75$$

#### Rule 4

If:  $210 \leq AG_{raw} < 260$

$$PM_{2.5} = [0.69(AG_{raw}/50 - 21/5) + 0.786(1 - (AG_{raw}/50 - 21/5))]AG_{raw} - 0.0862RH_{raw}(1 - (AG_{raw}/50 - 21/5)) + [2.966(AG_{raw}/50 - 21/5)] + [5.75(1 - (AG_{raw}/50 - 21/5))] + 8.84 \times 10^{-4}AG_{raw}^2(AG_{raw}/50 - 21/5)$$

#### Rule 5

If:  $AG_{raw} \geq 260$

$$PM_{2.5} = 2.966 + 0.69AG_{raw} + 8.84 \times 10^{-4}AG_{raw}^2$$

Where:

- $AG_{raw}$ — Raw PM<sub>2.5</sub> value ( $\mu\text{g}/\text{m}^3$ ) measured by the low-cost sensor
- $RH_{raw}$ — Relative humidity measurement (%)
- $PM_{2.5}$ — EPA-corrected PM<sub>2.5</sub> value ( $\mu\text{g}/\text{m}^3$ )

The original EPA PMS5003 correction algorithm documentation is provided in Appendix A.

The EPA correction algorithm uses both raw PM<sub>2.5</sub> and relative humidity measurements to compensate for environmental effects on optical particle sensor observations. By applying piecewise empirical correction equations, the method improves sensor accuracy relative to raw measurements and accounts for humidity effects and nonlinear sensor behavior across different PM<sub>2.5</sub> concentration ranges. However, because the EPA correction is based on fixed empirical relationships, it cannot adapt to local environmental conditions or capture more complex nonlinear interactions among PM<sub>2.5</sub> concentration, humidity, temperature, pressure, and spatial factors.

## 2.6 Machine Learning for Sensor Calibration

The use of machine learning in environmental monitoring applications is increasing in importance since it can effectively identify complex patterns in large datasets. Several machine learning approaches have been investigated in recent years for the calibration of low-cost air quality sensors. While classic correction equations rely on predefined coefficients, machine learning models are able to adapt to different environmental conditions and learn nonlinear correlations directly from the data [12],[13],[22].

The calibration process generally involves training a predictive model using low-cost sensor measurements together with co-located reference station measurements [12],[13]. Input features typically include environmental variables such as humidity, temperature, atmospheric pressure, and time information, as these features have a major impact on sensor performance. The trained model then outputs predicted corrected PM<sub>2.5</sub> values that are closer to the readings obtained from scientific reference instrumentation.

In the previous studies on the calibration of air quality, several machine learning techniques have been used. The simplest models are linear regression models, and they are often used as baseline methods. Linear regression is computationally efficient and easy to understand, but it is usually not capable of capturing the complex nonlinear relationships between environmental variables and sensor responses.

Tree-based machine learning models such as Random Forest and Gradient Boosting have demonstrated improved calibration performance in many studies [12],[13]. These methods can model nonlinear relationships and interactions more effectively than traditional regression techniques. XGBoost and LightGBM have also been widely applied because of their high predictive accuracy and efficient training performance [12],[13].

Previous calibration studies have reported substantial performance improvements when machine learning methods are applied to low-cost PM<sub>2.5</sub> sensor calibration. For example, Zimmerman et al. [12] used a Random Forest calibration model and achieved an  $R^2$  value of approximately 0.91 compared to approximately 0.72 obtained using simpler calibration approaches. Similarly, Esposito et al. [13] demonstrated that neural network-based calibration significantly reduced prediction errors compared with conventional calibration methods under varying environmental conditions. Field evaluations conducted by Zheng et al. [14] further showed that machine learning-based calibration methods consistently improved agreement between low-cost sensors and regulatory-grade reference instruments. These findings suggest that machine learning approaches can substantially improve calibration accuracy by better capturing nonlinear relationships between environmental variables and sensor measurements.

Deep learning approaches have also been applied to low-cost PM<sub>2.5</sub> sensor calibration. For example, Esposito et al. [13] evaluated artificial neural network (ANN) models using air quality sensor data collected under varying environmental conditions and demonstrated improved agreement between low-cost sensors and reference monitoring instruments compared with conventional calibration approaches. The study showed that ANN models were able to capture nonlinear relationships between particulate matter measurements and meteorological variables, leading to reduced prediction errors. Similar studies have reported that deep learning models can effectively utilize time-aligned observations, including low-cost sensor measurements, reference

PM<sub>2.5</sub> concentrations, temperature, and humidity, to improve calibration performance. However, these models generally require larger training datasets, increased computational resources, and more extensive hyperparameter tuning compared with simpler machine learning approaches. Consequently, while deep learning methods can achieve high predictive accuracy, model interpretability often becomes more challenging.

Support Vector Machines (SVMs) have also been successfully applied to calibrate PM<sub>2.5</sub>. SVM-based models can provide good performance in high-dimensional spaces but may become computationally expensive for very large datasets.

The capacity to incorporate multiple environmental factors simultaneously is one of the main advantages of machine learning-based calibration. Low-cost PM sensors are highly sensitive to temperature and humidity. For example, hygroscopic particle growth caused by high humidity can lead optical sensors to overestimate particulate matter concentrations [4],[6]. Machine learning models can automatically learn these environmental interactions from time-aligned observations containing low-cost sensor PM<sub>2.5</sub> measurements, reference monitoring station PM<sub>2.5</sub> measurements, relative humidity, temperature, atmospheric pressure, and other relevant environmental variables, eliminating the need for manually derived correction formulas.

Another benefit of machine learning-based calibration is adaptability. Conventional calibration equations are often derived under specific environmental conditions and may not generalize well across different deployment settings. Low-cost PM<sub>2.5</sub> sensor measurements can be influenced by factors such as relative humidity, temperature, smoke events, dust episodes, particle composition, seasonal variation, local emission sources, and sensor aging [3],[4],[6]. Machine learning models can learn complex relationships between these factors and sensor measurements directly from data, enabling more flexible calibration. However, the performance of a calibration model may still depend on the characteristics of the training data, including the sensor type, city, climate, deployment period, and evaluation setup used during model development [12],[22].

However, machine learning-based calibration also presents several challenges. Previous studies have shown that calibration performance can depend on the characteristics of the training data, including the sensor type, deployment location, climate, and monitoring period used during model development [12],[13]. As a result, models developed for one deployment scenario may require retraining or additional validation before being applied to different environmental conditions. In addition, highly complex machine learning models may be more difficult to interpret than traditional calibration approaches. Another practical challenge is the requirement for large quantities of high-quality co-located measurements from both low-cost sensors and reference monitoring stations to support model training and validation.

Despite these challenges, machine learning-based calibration has shown significant potential for improving the reliability of low-cost PM<sub>2.5</sub> sensors. In particular, the Linear Tree Regressor is well suited to sensor calibration because it combines nonlinear decision-tree partitioning with local linear regression models. This structure allows the model to capture complex environmental effects while maintaining interpretable linear relationships within each partition. By simultaneously utilizing low-cost sensor PM<sub>2.5</sub> measurements together with environmental variables such as relative humidity, temperature, and atmospheric pressure, the model can learn calibration relationships that are difficult to represent using fixed correction equations. This

thesis therefore investigates the potential of the Linear Tree Regressor for improving PM<sub>2.5</sub> calibration accuracy using a large-scale real-world urban monitoring dataset.

## 2.7 Previous Machine Learning Calibration Studies

Previous studies have investigated a variety of machine learning approaches for low-cost PM<sub>2.5</sub> sensor calibration, including linear regression, Random Forest, Gradient Boosting, Support Vector Regression (SVR), and Artificial Neural Networks (ANNs) [11],[12],[13],[14]. These studies generally reported improved calibration performance compared with traditional empirical correction formulas because machine learning methods can model nonlinear relationships between sensor measurements and environmental variables more effectively.

However, the reported studies differ substantially in terms of sensor technology, deployment location, monitoring period, calibration methodology, and evaluation metrics. Some studies focused on short-term co-location experiments conducted at a single monitoring site, while others evaluated sensor networks deployed across different urban environments. In addition, different studies used different low-cost sensor types, target pollutants, environmental variables, preprocessing procedures, and performance metrics such as MAE, RMSE, and R<sup>2</sup>. As a result, direct comparison of calibration performance across studies is challenging.

Many previous investigations were also conducted within a single city or monitoring network, making it difficult to assess how calibration models perform under varying environmental and geographical conditions. Consequently, important research questions remain regarding the spatial robustness, scalability, and transferability of machine learning-based calibration methods. Furthermore, relatively few studies have investigated how the number of reference monitoring stations influences calibration performance or explicitly evaluated the relationship between sensor-reference station distance and calibration accuracy. These gaps motivate the present study.

Table 2. Examples of machine learning calibration performance reported in previous studies

| Study                        | Model                 | Reported Performance                           |
|------------------------------|-----------------------|------------------------------------------------|
| Zimmerman et al. (2018) [12] | Random Forest         | $R^2 \approx 0.91$                             |
| Esposito et al. (2016) [13]  | Neural Network        | Significant reduction in calibration error     |
| Zheng et al. (2018) [14]     | ML Calibration Models | Improved agreement with reference measurements |
| Barkjohn et al. (2021) [11]  | EPA Correction        | $R^2 \approx 0.69$                             |

## 2.8 Linear Tree Regressor

The Linear Tree Regressor is a hybrid machine learning model that combines the hierarchical partitioning capability of decision trees with the predictive power of linear regression models [15]. The model fits distinct linear regression equations into the decision tree's terminal nodes rather than generating constant prediction values within each tree leaf.

First, the feature space is divided into smaller regions by using decision tree splitting rules. Then, train a local linear regression model on each partition to capture the relationships between input variables and the target variable. This structure enables the model to capture nonlinear global patterns and locally linear relationships in the data.

One of the major advantages of the Linear Tree Regressor is that it is able to capture complicated nonlinear interactions, while still maintaining fairly high interpretability compared to more sophisticated machine learning methods such as deep neural networks [15]. Moreover, the model usually has lower overfitting tendencies compared with fully grown decision trees, since the local linear models can better explain the variations within each partition.

The Linear Tree Regressor is particularly suitable for PM<sub>2.5</sub> calibration, because the behavior of low-cost particulate matter sensors is strongly affected by environmental conditions (e.g., humidity, temperature, and atmospheric pressure). These environmental effects often produce nonlinear sensor-response patterns that cannot be fully captured using fixed empirical correction formulas or simple linear regression approaches.

## 2.9 Spatial Calibration and Distance Effects

Previous research has shown that the accuracy of low-cost sensor calibration can be influenced by the spatial distance between low-cost sensors and reference monitoring stations [8],[9]. Air pollution concentrations may vary significantly across urban environments because of traffic density, industrial activity, meteorological conditions, and local urban morphology. Consequently, nearby reference stations may provide environmental conditions that more accurately represent the local pollution characteristics observed by low-cost sensors.

In contrast, the calibration based on distant reference monitoring stations may bring in the spatial mismatch effects, in which the differences of local environmental and pollution conditions decrease the calibration accuracy. Several studies have therefore emphasized the importance of spatial considerations when designing large-scale air quality monitoring networks and calibration frameworks [8],[9],[23].

Despite growing interest in spatial calibration methods, several important research gaps still remain. These include determining the optimal density and location of reference monitoring stations, identifying practical distance thresholds for effective calibration, exploring the transferability of calibration across multiple cities and investigating the spatial robustness of machine learning-based calibration models under different environmental conditions.

## 2.10 Research Gap

Although previous studies have demonstrated the potential of machine learning methods for low-cost PM<sub>2.5</sub> sensor calibration, several important limitations remain in the existing literature. Many studies rely on relatively small datasets collected from short-term co-location experiments involving only a limited number of sensors and reference monitoring stations. As a result, the scalability and robustness of many proposed calibration models remain insufficiently evaluated.

In addition, numerous calibration studies focus primarily on improving prediction accuracy while giving limited attention to spatial distance effects between low-cost sensors and reference stations. Research investigating the optimal number and placement of reference monitoring stations for large-scale urban calibration frameworks is also relatively limited.

Moreover, many existing machine learning calibration approaches utilize complex black-box models that are computationally expensive and difficult to interpret. While such models may achieve high predictive performance, their lack of transparency can reduce trust, practical

usability, and deployment feasibility for environmental monitoring applications. Relatively limited research has investigated calibration approaches that simultaneously provide strong predictive performance, interpretability, and model simplicity.

Furthermore, many existing studies do not perform large-scale cross-validation experiments across different sensor-reference station combinations, making it difficult to evaluate the generalizability and stability of the proposed models under varying environmental conditions.

This thesis addresses these limitations by leveraging a large real-world air quality dataset containing millions of observations from multiple low-cost sensors and reference monitoring stations. The proposed framework also incorporates spatial distance analysis, reference station optimization experiments, stratified large-scale cross-validation, and statistical evaluation methods to investigate calibration robustness and scalability. In addition, the study evaluates an interpretable and computationally efficient Linear Tree Regressor model capable of balancing predictive performance with model transparency and practical deployment suitability for urban air quality monitoring systems.

## 2.11 Summary

In this section, a review of the previous research work on air pollution monitoring, PM<sub>2.5</sub> health impacts, conventional reference monitoring systems, low-cost air quality sensors, EPA correction methods, machine learning-based calibration techniques, and spatial calibration analysis was presented. The review discussed the strengths and limitations of the state-of-the-art calibration approaches and also pointed out several important research gaps in terms of spatial robustness, reference station optimization, large-scale validation, and model generalization.

## 3. Research Methodology

### 3.1 Introduction

This section describes the overall research methodology, experimental design, machine learning framework, evaluation metrics, and statistical analysis methods used in this thesis.

### 3.2 Research Workflow

The research workflow consisted of the following stages.

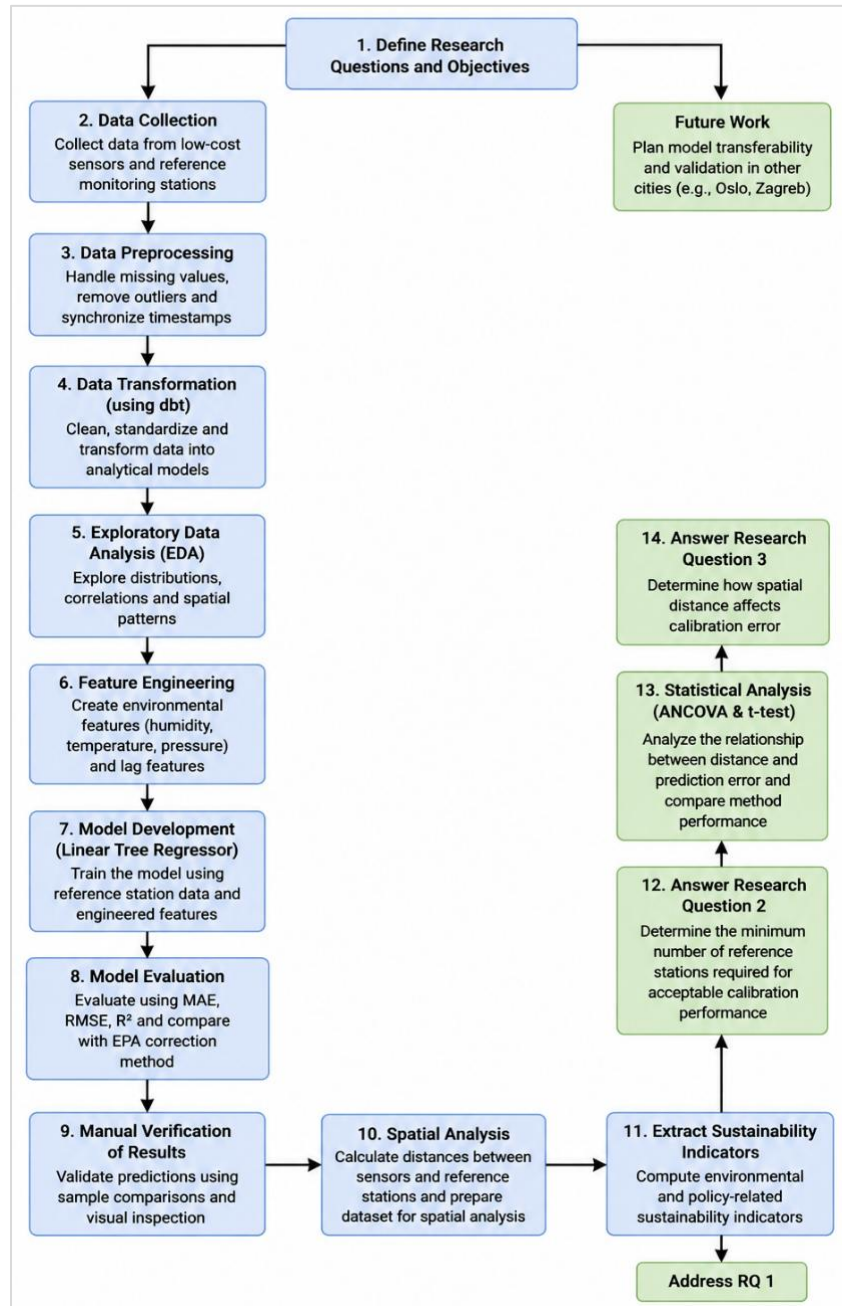


Figure 1. Research Workflow

### 3.3 Dataset Description and Preprocessing

This study used the SensEURLCity dataset [1]. It is a large open-access air quality dataset collected from three European cities, Antwerp (Belgium), Oslo (Norway), and Zagreb (Croatia) during 2020 – 2021 (See *Figure 2*). The dataset was developed within the AirSensEUR project coordinated by the Joint Research Centre (JRC) of the European Commission.

The main objective of the SensEURLCity project was to evaluate the performance of low-cost air quality sensors under different environmental and meteorological conditions and to support the development of calibration methods for urban air pollution monitoring.

The complete dataset contains measurements collected from 85 AirSensEUR low-cost sensor systems deployed across the three cities. 34 sensor systems in Antwerp, 34 sensor systems in Oslo and 17 sensor systems in Zagreb.

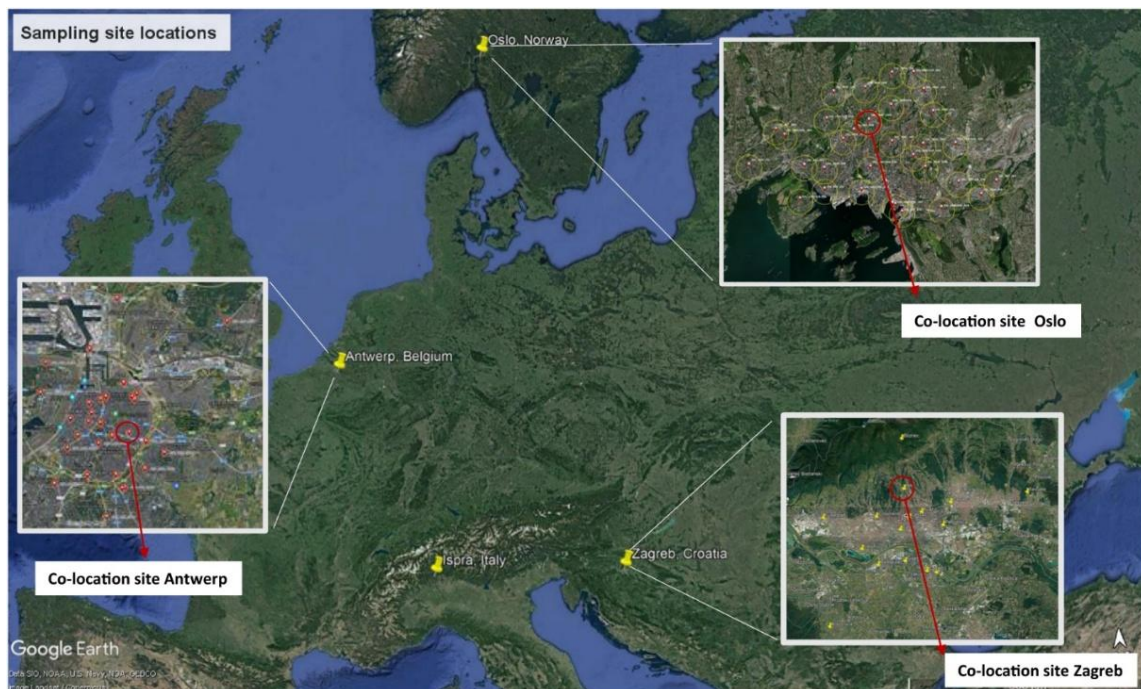


Figure 2. Deployment locations of the SensEURLCity low-cost air quality monitoring campaigns conducted in Antwerp, Oslo, and Zagreb

Source: Adapted from Van Poppel et al. [1]

The Antwerp deployment was selected for this study as it provided the most suitable environment for large-scale PM<sub>2.5</sub> calibration analysis. Antwerp included 34 AirSensEUR sensor systems distributed over a wide spatial distribution with multiple reference monitoring stations and dedicated urban traffic and background locations. In addition, the Antwerp dataset included long periods of co-location and deployment (See 3.3.3 Study Period) with very detailed PM<sub>2.5</sub> reference measurements, a wide environmental variability and continuous monitoring over a long time. These features facilitated a more reliable evaluation of calibration performance, spatial distance effects and reference station optimization in a consistent urban environment.

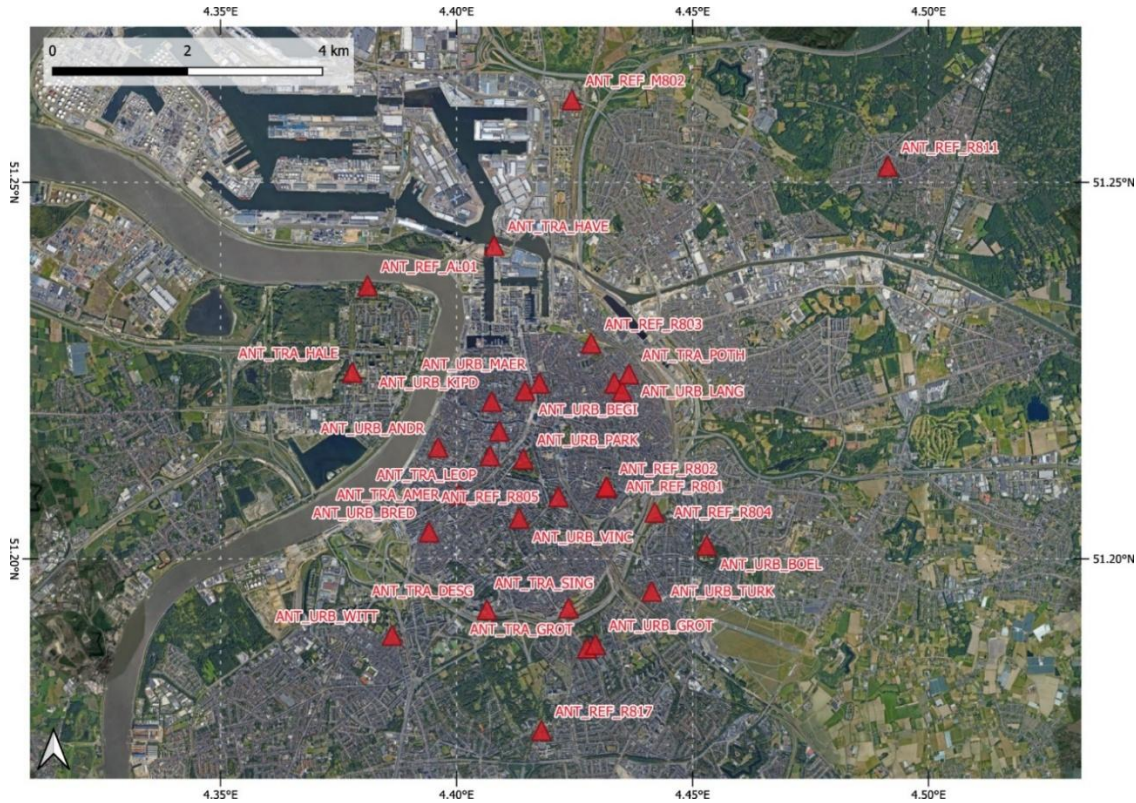


Figure 3. Spatial distribution of low-cost sensors and reference monitoring stations within the Antwerp deployment used in this thesis.

Source: Adapted from Van Poppel et al. [1]

### 3.3.1 AirSenseEUR Sensor Platform

The AirSenseEUR platform is an open low-cost air quality sensing system designed to monitor both pollutant concentrations and meteorological conditions [1],[2]. The sensor platform includes sensors for PM<sub>1</sub>, PM<sub>2.5</sub>, PM<sub>10</sub>, NO, NO<sub>2</sub>, O<sub>3</sub>, CO, CO<sub>2</sub>, temperature, relative humidity, and atmospheric pressure.

Particulate matter was measured with a variety of sensor configurations including 5325CAT, 5325CST, and OPCN3 optical particle counters. The 5325CAT measured PM<sub>10</sub>, PM<sub>2.5</sub>, and PM<sub>1</sub> concentrations calibrated by the manufacturer and reported in µg/m<sup>3</sup> using proprietary calibration algorithms based on particle-bin counts. The 5325CST measured uncorrected concentrations of PM<sub>10</sub>, PM<sub>2.5</sub>, and PM<sub>1</sub> based on particle bin counts with an assumed particle density and without manufacturer calibration.

The study focused specifically on PM<sub>2.5</sub> calibration because fine particulate matter is one of the most harmful air pollutants affecting human health.

### 3.3.2 Raw Dataset

The Antwerp deployment dataset consisted of 34 CSV (Comma-Separated Values) files with a total size of approximately 10.3 GB. Each CSV file contained measurements collected from a single AirSenseEUR low-cost sensor system deployed during the Antwerp monitoring campaign.

The raw dataset was initially imported into DuckDB using a Python-based ingestion pipeline. The imported table, named *antwerp\_measurements*, contained 92 columns, including timestamp information, geolocation coordinates (latitude and longitude), location identifiers, meteorological variables, quality-control flag columns, particulate matter measurements, reference station measurements, file names, and sensor identifiers (ASE\_Id). After import, the table contained 18,757,657 rows of minute-level observations.

DuckDB was selected because it provides efficient analytical processing for large-scale time-series datasets while requiring relatively low system resources.

The dataset included particulate matter measurements obtained from two sensor technologies: the Plantower PMS5003 sensor and the Alphasense OPC-N3 optical particle counter. Within the AirSensEUR platform, the PMS5003 measurements were available in two forms: CAT and CST. The CAT variables represent the manufacturer-calibrated PMS5003 particulate matter measurements, whereas the CST variables represent raw PMS5003 particle measurements before manufacturer calibration. The OPC-N3 is a separate optical particle counter that independently estimates particulate matter concentrations. All three measurement streams included PM<sub>1</sub>, PM<sub>2.5</sub>, and PM<sub>10</sub> variables.

In this study, the CAT PM<sub>2.5</sub> measurements were selected for the primary calibration experiments. Exploratory analysis showed that the manufacturer-calibrated PMS5003 PM<sub>2.5</sub> measurements exhibited closer agreement with the reference monitoring station observations than the corresponding CST and OPC-N3 PM<sub>2.5</sub> measurements. The detailed sensor selection procedure is described in Section 3.3.6.

The dataset also contained regulatory-grade reference measurements (Ref.PM<sub>1</sub>, Ref.PM<sub>2.5</sub>, Ref.PM<sub>10</sub>) from co-located reference monitoring stations. The reference PM<sub>2.5</sub> measurements were used as the ground-truth target variable for training and evaluating the machine learning calibration models developed in this study.

During the exploratory data analysis phase, it was identified that three reference monitoring station locations (REF\_R801, REF\_R802, and REF\_R804) each contained two co-located low-cost sensor systems operating during the same monitoring period. In this study, a duplicated deployment location refers to a reference monitoring station where two low-cost sensor systems were installed at the same location and measured the same environmental conditions simultaneously. Retaining both sensors would introduce redundancy into the calibration dataset because both systems were exposed to nearly identical environmental and pollution conditions.

To address this issue, a duplicate-sensor selection procedure was performed. First, the duplicated sensor pairs were identified at REF\_R801, REF\_R802, and REF\_R804. Second, the PM<sub>2.5</sub> measurements from each low-cost sensor were compared against the corresponding reference monitoring station PM<sub>2.5</sub> observations. Calibration performance was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which directly measure prediction error in  $\mu\text{g}/\text{m}^3$  and therefore provide an intuitive basis for sensor comparison. For each duplicated pair, the sensor exhibiting lower MAE and RMSE values was retained, while the poorer-performing duplicate sensor was excluded from further analysis.

The results of this comparison are presented in Table 3. At REF\_R801, sensor 40499C ( $MAE = 5.38$ ,  $RMSE = 7.65$ ) outperformed sensor 4043B1 ( $MAE = 6.58$ ,  $RMSE = 9.07$ ). At REF\_R802,

sensor 4049A6 ( $MAE = 5.55$ ,  $RMSE = 7.69$ ) outperformed sensor 4043A7 ( $MAE = 7.50$ ,  $RMSE = 10.56$ ). Similarly, at REF\_R804, sensor 40499F ( $MAE = 7.39$ ,  $RMSE = 10.46$ ) performed better than sensor 4043AE ( $MAE = 7.97$ ,  $RMSE = 11.80$ ). Consequently, sensors 4043B1, 4043A7, and 4043AE were excluded, while sensors 40499C, 4049A6, and 40499F were retained for subsequent analyses. This procedure reduced redundancy and ensured that only the best-performing sensor from each duplicated deployment location was included in the calibration dataset.

Table 3. Duplicate Sensor Selection Results Based on  $PM_{2.5}$  Agreement with Reference Monitoring Stations

| Location ID     | Sensor Id     | MAE         | RMSE         |
|-----------------|---------------|-------------|--------------|
| REF_M802        | 4065EA        | 5.53        | 7.65         |
| <b>REF_R801</b> | <b>4043B1</b> | <b>6.58</b> | <b>9.07</b>  |
| <b>REF_R801</b> | <b>40499C</b> | <b>5.38</b> | <b>7.65</b>  |
| REF_AL01        | 4065DA        | 5.22        | 7.32         |
| <b>REF_R802</b> | <b>4043A7</b> | <b>7.50</b> | <b>10.56</b> |
| <b>REF_R802</b> | <b>4049A6</b> | <b>5.55</b> | <b>7.69</b>  |
| REF_R803        | 4067BD        | 7.05        | 9.65         |
| <b>REF_R804</b> | <b>4043AE</b> | <b>7.97</b> | <b>11.80</b> |
| <b>REF_R804</b> | <b>40499F</b> | <b>7.39</b> | <b>10.46</b> |
| REF_R805        | 4067B3        | 5.65        | 7.76         |
| REF_R811        | 40642B        | 7.31        | 10.20        |
| REF_R817        | 4047D7        | 7.56        | 10.66        |

MAE and RMSE were selected as the primary criteria because they directly quantify prediction error in  $PM_{2.5}$  concentration units ( $\mu g/m^3$ ), making them more suitable for comparing sensor agreement with reference measurements.

### 3.3.3 Study Period

To ensure temporal consistency across all sensor systems, only measurements collected between **2020-06-29 12:20:00** and **2021-02-17 07:18:00** were included in the study (see *Figure 4*). This common deployment period corresponds to the time interval during which all selected low-cost sensors and reference monitoring stations were operating simultaneously.

The common study period was applied consistently across all experiments, including the EPA correction comparison (RQ1), Linear Tree Regressor model training and evaluation using stratified five-fold cross-validation (RQ1), the reference station subset analysis (RQ2), and the spatial distance analysis involving regression analysis and ANCOVA (RQ3). Restricting the dataset to this shared monitoring period ensured that all calibration models were trained and evaluated using temporally aligned observations collected under comparable environmental conditions.

The start timestamp corresponds to the deployment time of sensor 408168, which was the last selected low-cost sensor to become operational. The end timestamp corresponds to the removal

time of sensor 40642B, which was the first selected low-cost sensor to be removed from operation. Consequently, these timestamps define the start and end of the common study period used after filtering, ensuring that all selected low-cost sensors and reference monitoring stations were operating simultaneously throughout the analysis period.



Figure 4. Deployment timelines of low-cost sensor systems in Antwerp.

Source: Adapted from Van Poppel et al. [1]

In *Figure 4*, each horizontal bar represents the operational history of a low-cost sensor system identified by its sensor ID (y-axis). The colored segments indicate the different deployment phases of the AirSensEUR campaign: Phase 1 (red) represents the feasibility study, Phase 2 (yellow) the pilot study, Phase 3 (green) the first co-location period, Phase 4 (blue) the deployment period, and Phase 5 (purple) the second co-location period. The left red vertical line marks the deployment time of sensor 408168, which was the last selected sensor to become operational. The right red vertical line marks the removal time of sensor 40642B, which was the first selected sensor to be removed from operation. Measurements collected between these two timestamps define the common study period used after filtering, ensuring that all selected low-cost sensors and reference monitoring stations were operating simultaneously during model training, EPA comparison, reference station optimization, and distance analysis.

### 3.3.4 Data Transformation Using dbt

A modular and reproducible data transformation pipeline was implemented using dbt (data build tool) on top of the DuckDB analytical database. The raw Antwerp measurements were initially stored in a DuckDB table named *antwerp\_measurements*, which contained approximately 18.7 million minute-level observations collected from all deployed AirSensEUR sensor systems in Antwerp. The purpose of the dbt pipeline was to systematically clean, standardize, transform, and prepare the raw sensor measurements for machine learning calibration experiments.

We have broken down the dbt workflow into several staging and mart layers. This layered design allowed for separation of concerns between raw data ingestion, preprocessing logic, EPA

correction calculations, hourly aggregation, and analytical model preparation. We implemented each transformation stage as a separate SQL model, making the workflow modular, visible, maintainable and repeatable.

The *stg\_antwerp\_measurements* model implemented the first transformation layer. This staging view regularized the raw dataset structure and simplified the downstream processing. The model renamed technical sensor variables into more readable column names, selected only variables needed for PM<sub>2.5</sub> calibration experiments and regularized timestamp formatting. Meteorological variables such as relative humidity, temperature and atmospheric pressure were also extracted and collocated with PM<sub>2.5</sub> observations. The staging process additionally retained important metadata including sensor identifiers, location identifiers, latitude, longitude, and quality-control flag columns.

After staging, the *mart\_antwerp\_measurements* analytical view was created. This model filtered the dataset according to the common study period used throughout the thesis experiments. After the study-period filtering step, the resulting analytical views contained 11,237,712 minute-level observations.

The next group of transformations focused on PM<sub>2.5</sub> calibration datasets for Plantower PMS5003 CAT sensor measurements. The *mart\_pm25\_corrected\_CAT* model filtered observations according to multiple quality constraints. First, only observations with valid reference PM<sub>2.5</sub> measurements were retained. Second, PM<sub>2.5</sub> quality-control flags were evaluated to exclude invalid or unreliable measurements. Third, only records collected during the defined study period were preserved. This model therefore created a cleaned PM<sub>2.5</sub> dataset suitable for calibration experiments using co-located reference station measurements.

The *mart\_pm25\_epa\_corrected\_CAT* model extended the previous transformation by applying the EPA PM<sub>2.5</sub> correction formula to raw low-cost sensor measurements. The EPA correction algorithm used raw PM<sub>2.5</sub> concentrations together with relative humidity measurements to estimate corrected PM<sub>2.5</sub> values. Only rows containing non-null low-cost sensor PM<sub>2.5</sub> values were retained during this step. The EPA-corrected values generated in this model were later used as the baseline calibration approach for comparison against the machine learning models.

Following EPA correction, the *mart\_pm25\_epa\_corrected\_hourly\_CAT* model aggregated minute-level measurements into hourly observations using the median value within each hourly window. Hourly aggregation was selected to reduce measurement noise and improve temporal stability. The model included both raw and EPA-corrected PM<sub>2.5</sub> values together with reference station measurements. The hourly aggregation step also simplified cross-validation experiments and reduced computational complexity during model training.

A parallel transformation workflow was implemented for CAT sensor measurements that specifically retained low-cost sensor observations independently from reference co-location filtering. The *mart\_pm25\_corrected\_CAT\_LCS* model filtered the dataset using study-period constraints, low-cost sensor PM<sub>2.5</sub> availability, and PM<sub>2.5</sub> quality-control flags. This dataset was used to preserve valid low-cost sensor observations that could later be paired with multiple reference monitoring stations.

The *mart\_pm25\_epa\_corrected\_CAT\_LCS* model used the EPA correction equations on the CAT low-cost sensor measurements. The same filtering constraints were applied to keep only

valid PM<sub>2.5</sub> observations. Finally, the final hourly analytical dataset for CAT low-cost sensors was produced with the *mart\_pm25\_epa\_corrected\_hourly\_CAT\_LCS* model. This transformation converted the data to hourly medians and maintained the raw PM<sub>2.5</sub> measurements, the EPA-corrected PM<sub>2.5</sub> values, meteorological variables, and the sensor-reference pairing information.

In addition to PMS5003 CAT sensors, a separate transformation workflow was implemented for OPCN3 optical particle counter measurements. The *mart\_pm25\_lag\_corrected\_OPCN3* model filtered the OPCN3 dataset according to the study period, PM<sub>2.5</sub> quality-control flags, valid low-cost sensor measurements, and valid reference PM<sub>2.5</sub> observations.

The *mart\_pm25\_epa\_corrected\_OPCN3* model applied the EPA correction equations to OPCN3 PM<sub>2.5</sub> measurements. As in the CAT workflow, the observations from OPCN3 were aggregated to hourly data using the *mart\_pm25\_epa\_corrected\_hourly\_OPCN3* model. To increase robustness and consistency of the measurements we again applied hourly median aggregation.

The full transformation pipeline allowed several important preprocessing operations required in the thesis experiments. These included data cleaning, standardized naming conventions, study-period filtering, quality control filtering, removal of invalid observations, EPA correction calculation, hourly aggregation, and efficient analytical querying. The modular structure of dbt also ensured full reproducibility of the preprocessing workflow and simplified iterative experimentation during model development.

Another benefit of the dbt-based architecture was the ability to decouple preprocessing logic from the rest of the pipeline and expose it as re-usable analytical layers. Each transformation model was independently validated, tested and reusable for multiple ML experiments. This enhanced the maintainability and reduced the risk of inconsistencies across experiments.

The transformed analytical datasets generated by the dbt workflow formed the basis for all calibration, optimization, and spatial analysis experiments conducted in this thesis.

The final machine learning working table contained 1,989,004 observations representing pairwise spatial combinations between 9 reference monitoring stations and 31 low-cost sensor systems. The working table was constructed using CAT PM<sub>2.5</sub> measurements, regulatory-grade reference PM<sub>2.5</sub> observations, environmental variables (relative humidity, temperature, and atmospheric pressure), and spatial distance information. Only CAT PM<sub>2.5</sub> measurements were included in the final calibration dataset because exploratory analysis indicated that they exhibited better agreement with the reference monitoring station observations than the alternative PM<sub>2.5</sub> measurement streams. The detailed sensor evaluation and selection procedure is presented in Section 3.3.6.

The dataset was subsequently used for model training, cross-validation, EPA baseline comparison, and spatial statistical analysis.

Table 4. Data Description

| Variable                                  | Description                                                | Format/Raw units  |
|-------------------------------------------|------------------------------------------------------------|-------------------|
| <i>lcs_id</i>                             | Low-cost sensor location identifier                        | String            |
| <i>refs_id</i>                            | Reference station location identifier                      | String            |
| <i>lcs_ase_id</i>                         | Low-cost sensor identifier                                 | String            |
| <i>lcs_lat, lcs_lon, ref_lat, ref_lon</i> | Latitude, Longitude coordinate of sensor location          | Decimal number    |
| <i>pm25_lcs</i>                           | Raw PM <sub>2.5</sub> measurement from the low-cost sensor | µg/m <sup>3</sup> |
| <i>pm25_lcs_epa</i>                       | EPA-corrected PM <sub>2.5</sub> value                      | µg/m <sup>3</sup> |
| <i>pm25_refs</i>                          | Reference station PM <sub>2.5</sub> measurement            | µg/m <sup>3</sup> |
| <i>rel_humidity</i>                       | Ambient relative humidity                                  | %                 |
| <i>temperature</i>                        | Ambient temperature                                        | °C                |
| <i>pressure</i>                           | Atmospheric pressure                                       | hPa               |
| <i>distance</i>                           | Distance between sensor and reference station              | meters            |
| <i>timestamp</i>                          | Date and time of observation                               | datetime          |

### 3.3.5 Hourly Aggregation Method

Minute-level PM<sub>2.5</sub> measurements were aggregated into hourly observations using the median value.

Hourly aggregation was selected because reference station measurements are commonly analyzed hourly, noise reduction improves model stability and temporal alignment becomes easier [1],[11].

The final hourly analytical datasets contained 66,624 observations for the OPCN3 workflow where reference PM<sub>2.5</sub> measurements were available, 65,540 observations for the CAT sensor workflow with non-null reference PM<sub>2.5</sub> measurements, and 185,293 observations for the CAT low-cost sensor (LCS) workflow containing all valid low-cost sensor observations after preprocessing and hourly aggregation.

### 3.3.6 CAT and OPCN3 Sensor Evaluation and Selection

Previous research on low-cost PM<sub>2.5</sub> calibration has frequently used measurements obtained from the Plantower PMS5003 particle sensor because it generally demonstrates stronger agreement with regulatory-grade PM<sub>2.5</sub> observations than many alternative low-cost optical particle sensors [11],[12],[14]. Within the AirSenseEUR platform, PM<sub>2.5</sub> measurements from the PMS5003 sensor are available as CAT and CST variables. The CAT variables represent the manufacturer-calibrated PMS5003 measurements, whereas the CST variables represent the corresponding raw PMS5003 measurements. In addition, the EPA PM<sub>2.5</sub> correction algorithm used in this thesis was specifically developed for PMS5003-based sensors.

To determine the most suitable PM<sub>2.5</sub> measurement stream for calibration, an exploratory comparison was conducted using the processed hourly analytical datasets. This evaluation was performed before the sensor-reference pair generation process and used co-located measurements collected at the same reference monitoring station locations. PM<sub>2.5</sub> measurements from the CAT stream and the OPC-N3 optical particle counter were compared against PM<sub>2.5</sub> observations from

the reference monitoring stations. The objective was to identify the measurement stream that demonstrated the strongest agreement with the reference measurements and was therefore most appropriate for the subsequent calibration experiments.

For clarity, *Figure 5* presents manufacturer-calibrated CAT PM<sub>2.5</sub> measurements after preprocessing and filtering. EPA-corrected PM<sub>2.5</sub> values were not used in this comparison because the purpose of the analysis was to evaluate the suitability of the CAT and OPC-N3 measurement streams before applying calibration methods.

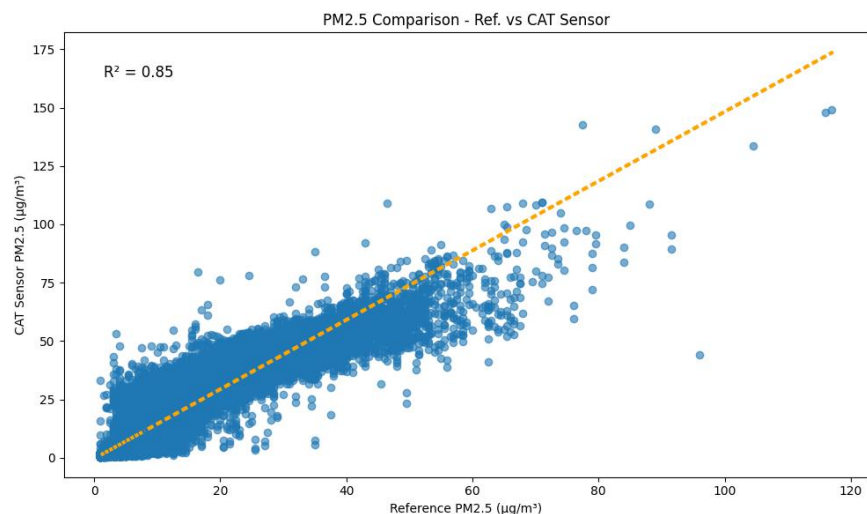


Figure 5. Scatterplot comparison between reference PM<sub>2.5</sub> measurements and low-cost PM<sub>2.5</sub> measurements obtained from CAT sensors after preprocessing and filtering

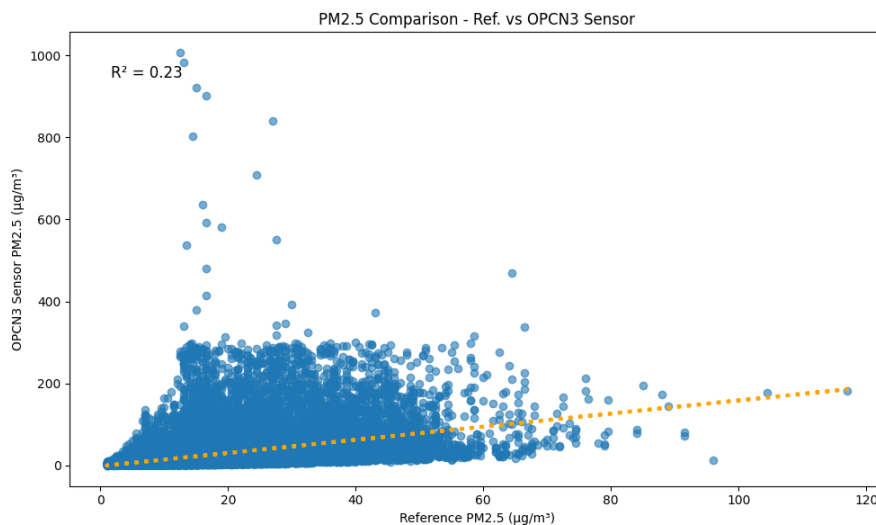


Figure 6. Scatterplot comparison between reference PM<sub>2.5</sub> measurements and low-cost PM<sub>2.5</sub> measurements obtained from OPCN3 sensors after preprocessing and filtering

The comparison results showed that CAT PM<sub>2.5</sub> measurements exhibited substantially stronger agreement with the reference observations than OPC-N3 measurements. As shown in *Table 5* and *Figure 5*, the CAT stream achieved a higher coefficient of determination ( $R^2 = 0.85$ ) and followed the reference measurement trend more consistently. In contrast, the OPC-N3 measurements displayed greater variability, wider dispersion, and more pronounced deviations

from the reference observations. Although increased scatter can be observed at higher  $PM_{2.5}$  concentrations, this pattern may also be influenced by environmental factors such as humidity, temperature, particle composition, and measurement uncertainty rather than sensor performance alone.

Table 5. Correlation and  $R^2$  comparison between CAT and OPCN3  $PM_{2.5}$  with Reference  $PM_{2.5}$  measurements

| Sensor Type  | Correlation (R) | $R^2$ Score |
|--------------|-----------------|-------------|
| CAT Sensor   | 0.92            | 0.85        |
| OPCN3 Sensor | 0.48            | 0.23        |

The observed behavior is consistent with findings reported in previous studies involving the SenseURCity dataset and Plantower PMS5003 sensors [1],[11]. Furthermore, the EPA  $PM_{2.5}$  correction algorithm used in this thesis was specifically designed for PMS5003-based measurements, providing additional justification for selecting the CAT stream. Based on the higher agreement with reference observations, the stronger  $R^2$  value, and the compatibility with the EPA correction framework, CAT  $PM_{2.5}$  measurements were selected as the primary low-cost sensor input for all subsequent machine learning calibration experiments.

### 3.4 Feature Selection

Four main input variables were selected:  $PM_{2.5}$  concentration, relative humidity, temperature and atmospheric pressure.

These features were selected because previous studies demonstrated strong environmental influence on PM sensor behavior [4],[6],[12].

### 3.5 EPA Correction Formula

The EPA  $PM_{2.5}$  correction algorithm described in Section 2.5 was implemented as the baseline calibration method in this study. The correction equations were applied to the raw  $PM_{2.5}$  measurements obtained from the 5325CAT low-cost sensors using the corresponding relative humidity observations.

The EPA correction calculations were implemented within the dbt transformation pipeline using SQL-based analytical models. The corrected  $PM_{2.5}$  values were then aggregated to hourly median observations and used as the baseline model for comparison with the Linear Tree Regressor calibration framework during the evaluation experiments.

Although the EPA correction method increases the accuracy of  $PM_{2.5}$  estimation compared to raw sensor measurements, it is a fixed empirical method, which cannot fully characterize the complex nonlinear interactions between environmental variables and sensor behavior. Thus, this study further explored whether machine learning methods can be used to achieve an improved calibration performance beyond the EPA baseline.

## 3.6 Machine Learning Model

### 3.6.1 Linear Tree Regressor

The primary machine learning model used in this study was the Linear Tree Regressor. It was selected because it offers a balance between predictive accuracy and interpretability [15], making it suitable for environmental calibration tasks and comparison with the EPA correction method.

Compared with more complex machine learning approaches such as Random Forests, XGBoost, and deep neural networks, the model is relatively transparent because each leaf node contains a standard linear regression equation. This structure enables a more straightforward interpretation of how environmental variables influence PM<sub>2.5</sub> predictions under different environmental conditions while maintaining the ability to model nonlinear relationships. Although more advanced models may achieve competitive performance, their evaluation is beyond the scope of this thesis and is suggested as future work.

The Linear Tree Regressor is a hybrid regression algorithm that combines the hierarchical partitioning capability of decision trees with the predictive power of linear regression models. Unlike traditional decision trees that output constant values at terminal nodes, the model fits a separate linear regression equation within each leaf node. This enables it to capture both nonlinear global patterns and localized linear relationships in the data [15].

The model operates in two main stages. First, the decision-tree component recursively partitions the feature space into smaller and more homogeneous regions based on the input variables. The splitting process identifies the optimal feature and threshold value that minimize prediction error at each node [15]. The splitting rule can be expressed as:

$$x_j \leq t$$

where  $x_j$  represents the selected feature and  $t$  represents the split threshold. In this study, the partitioning process used environmental and sensor-related variables such as raw PM<sub>2.5</sub> concentration, relative humidity, temperature, and atmospheric pressure.

In the Linear Tree Regressor, the split threshold ( $t$ ) is not manually specified. Instead, it is automatically determined during model training using the algorithm's optimization procedure. At each node, the model evaluates multiple candidate split thresholds across the available input features and selects the threshold that produces the greatest reduction in prediction error. Consequently, the threshold values are learned directly from the training data and may differ across tree nodes depending on the local characteristics of the dataset.

After the feature space is partitioned, a linear regression model is independently trained within each terminal leaf node. The local linear regression model predicts the corrected PM<sub>2.5</sub> concentration according to the following equation:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where  $\hat{y}$  represents the predicted PM<sub>2.5</sub> concentration,  $\beta_0$  is the intercept term,  $\beta_1, \beta_2, \dots, \beta_n$  are the regression coefficients, and  $(x_1, x_2, \dots, x_n)$  represent the input features. In this thesis, the target variable was the regulatory-grade reference PM<sub>2.5</sub> measurement obtained from co-located monitoring stations.

The Linear Tree Regressor was selected because low-cost PM<sub>2.5</sub> sensor measurements are influenced by complex nonlinear environmental effects. The effect of humidity and temperature on the sensor behavior may differ depending on the pollution conditions. Traditional linear regression models may not be able to capture these nonlinear interactions while standard decision trees may lead to unstable predictions and overfit noisy environmental data. The hybrid architecture of the Linear Tree Regressor can overcome these limitations by combining nonlinear partitioning with locally optimized linear models. This hybrid regression structure is based on recursive partitioning and localized linear modeling approaches described in previous machine learning research [15].

The model also provides smoother prediction behaviour compared to conventional regression trees. Standard regression trees generate piecewise constant outputs, which can create abrupt prediction changes near split boundaries. In contrast, the use of linear regression models within the leaf nodes allows continuous local prediction behaviour and improved generalization performance.

In this study, the Linear Tree Regressor was trained using stratified five-fold cross-validation to ensure robust performance evaluation across multiple sensor-reference station combinations. The model predictions were compared against the EPA correction baseline using Mean Absolute Error (*MAE*), Root Mean Squared Error (*RMSE*), and Coefficient of Determination (*R*<sup>2</sup>). The experimental results demonstrated that the Linear Tree Regressor significantly improved PM<sub>2.5</sub> calibration accuracy compared to the traditional EPA correction method.

The maximum depth of the implemented Linear Tree Regressor was set to 3, and the minimum number of samples in a leaf was 80. The base estimator in each leaf was a standard linear regression model. These hyperparameters were selected to optimize the predictive performance as well as the computational efficiency and the ability to generalize, while avoiding overfitting.

### 3.6.2 Model Inputs

The Linear Tree Regressor model used four environmental and sensor-related variables as input features. Those are raw PM<sub>2.5</sub> concentration obtained from the low-cost sensors, relative humidity, temperature, and atmospheric pressure. These variables were selected because previous studies have shown that low-cost PM<sub>2.5</sub> sensor measurements are strongly influenced by environmental conditions, particularly humidity and temperature.

The target variable used during model training was the regulatory-grade reference PM<sub>2.5</sub> measurement obtained from the co-located reference monitoring stations.

## 3.7 Cross-Validation Strategy

A stratified five-fold cross-validation strategy was implemented to evaluate the performance and robustness of the Linear Tree Regressor calibration model. Cross-validation was used to ensure that the reported results were reliable and not dependent on a single train-test split.

The proposed calibration framework was designed as a non-temporal prediction model. Each observation was treated independently, where the model received a single measurement record containing the low-cost sensor values, environmental variables, and spatial information to estimate calibrated PM<sub>2.5</sub> values. The model did not use temporal sequence information or lagged

historical observations. Therefore, the calibration process was conceptually similar to the EPA correction approach, which also calibrates each measurement independently using predefined correction relationships.

The dataset was divided into five equally sized folds while preserving the distribution of sensor-reference station combinations across all subsets. Stratification was performed using the combination of the reference monitoring station identifier and the low-cost sensor identifier. This approach ensured that each fold contained a balanced representation of different sensor-reference station pairs and environmental conditions.

During the cross-validation process, the model was trained five times independently. In each iteration, four folds (80% of the dataset) were used for training, and the remaining fold (20% of the dataset) was used for testing. The testing fold was rotated across iterations so that every observation was used exactly once for model evaluation.

The cross-validation framework provided a robust estimate of model performance by reducing the potential for bias associated with a single train-test split. It also enabled evaluation of the model's generalization ability across multiple sensor-reference station combinations and environmental conditions.

For each fold, calibration performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination ( $R^2$ ). These metrics were calculated on the test fold using observations that were not used during model training. The procedure generated one out-of-fold prediction for every observation in the dataset.

Final model performance was reported as the mean value and standard deviation across the five cross-validation folds, expressed as:

$$Performance = Mean \pm Standard Deviation$$

Therefore, all MAE, RMSE, and  $R^2$  values reported in RQ1 were derived from stratified five-fold cross-validation rather than from a single train-test split. This evaluation strategy provided a robust and statistically reliable comparison between the EPA correction baseline and the proposed machine learning calibration framework.

## 3.8 Evaluation Metrics

The performance of the calibration models was evaluated using three widely used regression metrics Mean Absolute Error ( $MAE$ ), Root Mean Squared Error ( $RMSE$ ), and Coefficient of Determination ( $R^2$ ). These metrics were selected because they provide complementary information about prediction accuracy, error magnitude, and model explanatory power.

### 3.8.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average absolute deviation of the predicted  $PM_{2.5}$  values from the reference  $PM_{2.5}$  observations.  $MAE$  gives a straightforward interpretation of the average prediction error in the same unit as the target variable ( $\mu g/m^3$ ).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where  $y_i$  represents the reference PM<sub>2.5</sub> value,  $\hat{y}_i$  represents the predicted PM<sub>2.5</sub> value, and  $n$  represents the total number of observations.

### 3.8.2 Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE) is the square root of the mean of the squared prediction errors. The *RMSE* is more sensitive to large prediction errors than the *MAE* and therefore useful for evaluating the robustness of models against extreme deviations.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Lower *MAE* and *RMSE* values indicate better calibration performance and smaller prediction errors.

### 3.8.3 Coefficient of Determination (R<sup>2</sup>)

The Coefficient of Determination (R<sup>2</sup>) measures the proportion of variance in the reference PM<sub>2.5</sub> observations that is explained by the calibration model. In this study, R<sup>2</sup> was calculated using the coefficient of determination implemented in the scikit-learn `r2_score` function. Higher R<sup>2</sup> values indicate better agreement between the predicted and reference PM<sub>2.5</sub> concentrations.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

where  $(y_i)$  represents the reference PM<sub>2.5</sub> observation,  $(\hat{y}_i)$  represents the predicted PM<sub>2.5</sub> value, and  $(\bar{y})$  represents the mean of the reference PM<sub>2.5</sub> observations.

Although R<sup>2</sup> is related to the squared Pearson correlation coefficient ( $r^2$ ), the two measures are not always identical. The coefficient of determination additionally accounts for prediction bias and is therefore more appropriate for evaluating regression-based calibration models. For this reason, R<sup>2</sup> was selected as the primary goodness-of-fit metric in this study.

All evaluation metrics (MAE, RMSE, and R<sup>2</sup>) were calculated on the test fold during stratified five-fold cross-validation. The final reported results correspond to the mean and standard deviation of the metric values obtained across the five folds.

## 3.9 Overall Accuracy Analysis

The first experiment evaluated whether the Linear Tree Regressor calibration model could improve PM<sub>2.5</sub> prediction accuracy compared to the EPA correction method. The evaluation framework compared the machine learning model against the EPA baseline under identical training and testing conditions.

A stratified five-fold cross-validation strategy was implemented to ensure robust and reliable model evaluation. Stratification was performed using the combined (reference station, low-cost sensor) pair identifier so that each fold contained a representative distribution of sensor-reference station combinations and environmental conditions.

During each cross-validation iteration, four folds were used for training and one-fold was used for testing, corresponding to approximately 80% training data and 20% testing data. To improve computational efficiency, random training subsampling was applied when the number of training observations exceeded 100,000 samples. The same testing observations were used for both the EPA correction method and the Linear Tree Regressor model to enable direct performance comparison.

Out-of-fold predictions were obtained and saved for each fold, for later spatial distance analysis. Calibration performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Coefficient of Determination ( $R^2$ ). Final performance values were reported as mean and standard deviation over all five folds.

After completing cross-validation, a final global Linear Tree Regressor model was trained using a representative subsample of the complete dataset. The resulting model structure was visualized to analyze the hierarchical feature partitioning and the linear regression equations generated within the leaf nodes.

### 3.10 Reference Station Optimization

The second experiment investigated how many reference monitoring stations were required to achieve reliable calibration performance. To evaluate this, subsets of reference stations ranging from one to nine stations were generated. For each subset size, random combinations of reference monitoring stations were selected in order to reduce computational complexity while still providing representative performance estimates.

For every generated subset, a new Linear Tree Regressor model was trained and evaluated using a common testing dataset. The calibration performance of each subset configuration was measured primarily using Mean Absolute Error (MAE).

The aim of this experiment was to analyze the relationship between the number of reference monitoring stations and the calibration accuracy. In particular, the analysis aimed to evaluate model scalability, infrastructure optimization, and the minimum number of reference stations to achieve acceptable calibration performance.

### 3.11 Spatial Distance Analysis and ANCOVA Evaluation

The third experiment investigated whether the spatial distance between low-cost sensors and reference monitoring stations influences calibration accuracy. This analysis was motivated by the practical deployment scenario considered in this thesis, where low-cost sensors and reference monitoring stations are often located at different positions within an urban monitoring network. As the separation distance increases, differences in local environmental conditions, emission sources, and pollutant concentrations may affect the similarity between sensor and reference measurements. Therefore, it is important to evaluate whether calibration performance deteriorates with increasing distance.

To investigate this relationship, prediction error was analyzed as a function of sensor-reference station distance. Simple linear regression was used to quantify the relationship between distance and prediction error and to estimate the rate at which calibration error changed with increasing distance. Analysis of Covariance (ANCOVA) was then applied to evaluate the effects of

distance, calibration method, and their interaction on prediction error. This approach enabled a comprehensive assessment of the spatial robustness of the Linear Tree Regressor relative to the EPA correction baseline.

### 3.11.1 Distance Computation

Distances between low-cost sensors and reference monitoring stations were calculated using their geographic coordinates, including latitude and longitude values. The haversine formula was used to estimate the great-circle distance between two locations on the Earth's surface while accounting for the Earth's curvature [8]. All distances were computed in kilometers and used in the subsequent spatial distance analysis and ANCOVA evaluation.

### 3.11.2 Error Calculation

The absolute prediction error was calculated to measure the difference between the predicted PM<sub>2.5</sub> values and the corresponding reference PM<sub>2.5</sub> observations. Absolute error was computed using the following equation:

$$\text{Absolute Error} = |y_i - \hat{y}_i|$$

Where  $y_i$  represents the reference PM<sub>2.5</sub> value and  $\hat{y}_i$  represents the predicted PM<sub>2.5</sub> value generated by the calibration model.

The same error calculation procedure was applied to both the EPA correction method and the Linear Tree Regressor model. The resulting absolute error values were subsequently used in the spatial distance analysis and ANCOVA evaluation to investigate how calibration accuracy changed with increasing sensor-reference station distance.

### 3.11.3 Distance Binning

To evaluate the change in calibration accuracy with increasing spatial distances, sensor-reference station pairs were grouped into distance intervals using a distance binning approach. The calculated distance values were divided into 20 equally spaced bins covering the entire range of observed sensor-reference station distances.

Each distance bin was defined as a particular range of distances, e.g. 0-1 km, 1-2 km, 2-3 km, etc. To further explore the relationship between spatial distances and calibration error, we binned the observations based on distances.

For each distance bin, the average Mean Absolute Error (MAE) was computed separately for the EPA correction method and the Linear Tree Regressor model. These accumulated error values were subsequently employed for regression analysis and graphical representation of distance-dependent calibration performance trends.

### 3.11.4 Linear Trend Analysis

Linear regression analysis was performed to evaluate how calibration error changed as the spatial distance between low-cost sensors and reference monitoring stations increased. The analysis was based on the average Mean Absolute Error (MAE) computed for each distance interval.

The fitted linear regression model was expressed as:

$$y = mx + b$$

Where  $y$  represents the average *MAE*,  $x$  represents the sensor-reference station distance,  $m$  represents the slope of the regression line, and  $b$  represents the intercept term.

The slope of the regression line was used to evaluate the spatial robustness of each calibration method. A smaller slope indicated that calibration error increased more slowly with distance, suggesting greater robustness to spatial separation between low-cost sensors and reference monitoring stations.

In addition, 95% confidence intervals were computed using the standard error of the mean to visualize uncertainty across the distance intervals and improve interpretation of the distance-related error trends.

### 3.11.5 ANCOVA Statistical Analysis

The association of calibration error with spatial distance was statistically tested with an ANCOVA model. ANCOVA was chosen because the analysis included a continuous independent variable (distance) and a categorical independent variable (calibration method).

The regression analysis presented in Section 4.4.2 was performed on binned mean errors for visualization purposes, whereas the ANCOVA analysis used observation-level absolute prediction errors to retain the full dataset and maximize statistical power.

The absolute prediction error was used as the dependent variable in the analysis. Independent variables included the spatial distance between the low-cost sensor and reference monitoring station, the calibration method (EPA correction or Linear Tree Regressor), and the interaction of distance and calibration method.

The ANCOVA model was formulated as:

$$Error = \beta_0 + \beta_1 Distance + \beta_2 Model + \beta_3 (Distance \times Model) + \epsilon$$

Where  $\beta_0$  represents the intercept term,  $\beta_1$  represents the effect of spatial distance,  $\beta_2$  represents the effect of calibration method, and  $\beta_3$  represents the interaction effect between distance and calibration method.

Cluster-robust standard errors were applied using the sensor-reference station pair identifier to account for repeated observations within the same sensor-reference pair. This approach improved the statistical reliability of the analysis because observations belonging to the same sensor-reference pair were not completely independent.

The ANCOVA analysis evaluated whether calibration error increased with spatial distance and whether the relationship between distance and prediction error differed between the EPA correction method and the Linear Tree Regressor model.

## 3.12 Tools and Technologies

Several software tools and analytical libraries were used throughout this study to support data processing, transformation, machine learning model development, statistical analysis, and visualization.

Data analysis and machine learning implementation were performed using Python as the primary programming language. Pandas and NumPy were used for data manipulation, preprocessing, and numerical computations. Scikit-learn was used for machine learning model development, cross-validation, and performance evaluation.

DuckDB was used for efficient analytical querying and storage of the large-scale time-series sensor dataset. In addition, the modular SQL-based data transformation pipeline, including preprocessing, filtering, EPA correction, and hourly aggregation, was implemented using dbt (data build tool).

The experiments and analyses were conducted in Jupyter Notebook, which provided an interactive environment for model development and visualization. Spatial distance calculations and regression visualizations were performed using Matplotlib, which was also used to generate figures and graphical analyses. Statistical analyses, including ANCOVA, and confidence interval calculations, were performed using SciPy and other scientific computing libraries.

The combination of Python, DuckDB, dbt, Jupyter Notebook, and scientific computing libraries provided an integrated environment for data processing, machine learning model development, statistical analysis, and visualization throughout the study.

### 3.13 Source Code Availability

The implementation source code, SQL transformation scripts, dbt models, and machine learning experiments developed for this thesis are publicly available in the following GitHub repositories:

<https://github.com/udayakkarunarathna/pm25-sensor-calibration-framework> ,  
<https://github.com/AnjanaLakshan/pm25-sensor-calibration-framework>

The repository includes the preprocessing pipeline, feature engineering workflow, model training scripts, evaluation procedures, and statistical analysis code used throughout this study [16].

### 3.14 Summary

This section presented the research methodology and experimental framework used in this thesis. The SensEURCity dataset, preprocessing workflow, dbt-based transformation pipeline, implementation of the EPA correction method, and the feature selection process were introduced. The section also described the calibration model based on the Linear Tree Regressor, the stratified cross-validation strategy, the experiments for reference station optimization, and the spatial distance analysis using regression analysis, and ANCOVA. These methodological components formed the basis for assessing the effectiveness of the proposed machine-learning-based post-hoc calibration framework for low-cost PM<sub>2.5</sub> sensors.

## 4. Results and Analysis

### 4.1 Introduction

This section presents the results and analysis of the experiments conducted in this thesis. The findings are organized according to the three research questions and include evaluation of the Linear Tree Regressor calibration framework, reference station optimization experiments, spatial distance analysis, and assessment of model generalization potential. The performance of the proposed machine learning approach is compared against the U.S. EPA correction baseline using multiple statistical evaluation metrics.

### 4.2 RQ1 – Comparison with EPA Correction Method

The first experiment investigated whether the Linear Tree Regressor model could improve PM<sub>2.5</sub> calibration accuracy compared to the standard EPA correction method. The evaluation was conducted using stratified five-fold cross-validation to ensure robust and reliable performance estimation across different sensor-reference station pairs.

The EPA correction method was selected as a baseline calibration approach because it is one of the most widely used post-processing methods for Plantower PMS5003 low-cost sensors. The EPA algorithm estimates corrected particulate matter concentrations using piecewise correction equations based on PM<sub>2.5</sub> measurements and relative humidity observations. Although the method accounts for changes in humidity, it relies on a fixed empirical formula with predefined coefficients. Consequently, the EPA correction cannot learn site-specific relationships from data or adapt its calibration behavior based on additional environmental variables such as temperature and atmospheric pressure.

The proposed machine learning framework employed a Linear Tree Regressor model trained on four environmental and sensor-related variables: raw PM<sub>2.5</sub> concentration, relative humidity, temperature and atmospheric pressure. The model was tested on the same training and testing folds used for the EPA baseline to ensure a fair comparison between the two calibration approaches.

The experiment results showed that the calibration performance was greatly enhanced with the proposed machine learning model. The Linear Tree Regressor yielded lower Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values and higher Coefficient of Determination ( $R^2$ ) values than the EPA correction method.

Table 6. Cross-Validation Performance Comparison

| Model                           | MAE  | MAE SD | RMSE | RMSE SD | $R^2$ | $R^2$ SD |
|---------------------------------|------|--------|------|---------|-------|----------|
| EPA correction                  | 3.55 | 0.004  | 5.29 | 0.017   | 0.69  | 0.002    |
| The Linear Tree Regressor model | 2.31 | 0.009  | 3.91 | 0.035   | 0.83  | 0.002    |

Although the EPA correction method achieved an  $R^2$  value of 0.69, this level of performance is consistent with previously reported EPA validation studies and still represents a substantial improvement over uncorrected low-cost sensor measurements. It should also be noted that the EPA performance reported in this experiment is not directly comparable to the sensor selection

results presented in Section 3.3.6. The sensor evaluation stage used co-located hourly measurements collected at reference monitoring station locations, where CAT measurements were compared directly with nearby reference observations and achieved an  $R^2$  of approximately 0.85. In contrast, the EPA evaluation in RQ1 was performed using the full machine learning working dataset containing approximately 1.99 million sensor-reference station pairs. These pairs include low-cost sensors and reference monitoring stations located at different locations across Antwerp, introducing additional spatial variability. Consequently, the EPA-corrected  $PM_{2.5}$  values were compared against reference  $PM_{2.5}$  observations from multiple stations at varying distances, making the prediction task substantially more challenging and resulting in lower overall  $R^2$  values.

The decrease in  $MAE$  suggests that the machine learning model produced predictions that were systematically closer to the reference monitoring station measurements. Similarly, the lower  $RMSE$  values indicate that the proposed model performed better in handling higher prediction deviations than the EPA correction method.

The improved performance is mainly due to the ability of the Linear Tree Regressor to account for nonlinear interactions between environmental variables and low-cost sensor behavior. Optical PM sensors are affected by environmental conditions such as humidity and temperature in a highly nonlinear way.

The decision-tree partitioning structure of the Linear Tree Regressor enabled the model to split the feature space into locally homogeneous regions by fitting separate linear regression models in each leaf node. This increased the flexibility and adaptability of the calibration process in different environmental conditions.

The global Linear Tree Regressor visualization shown in *Figure 7* was generated using a representative subset of 100,000 observations sampled from the full working dataset. The model divided the feature space according to  $PM_{2.5}$  concentration, relative humidity, temperature, and atmospheric pressure. Each terminal leaf node contained a separate local linear regression model used to estimate calibrated  $PM_{2.5}$  concentrations under different environmental conditions.

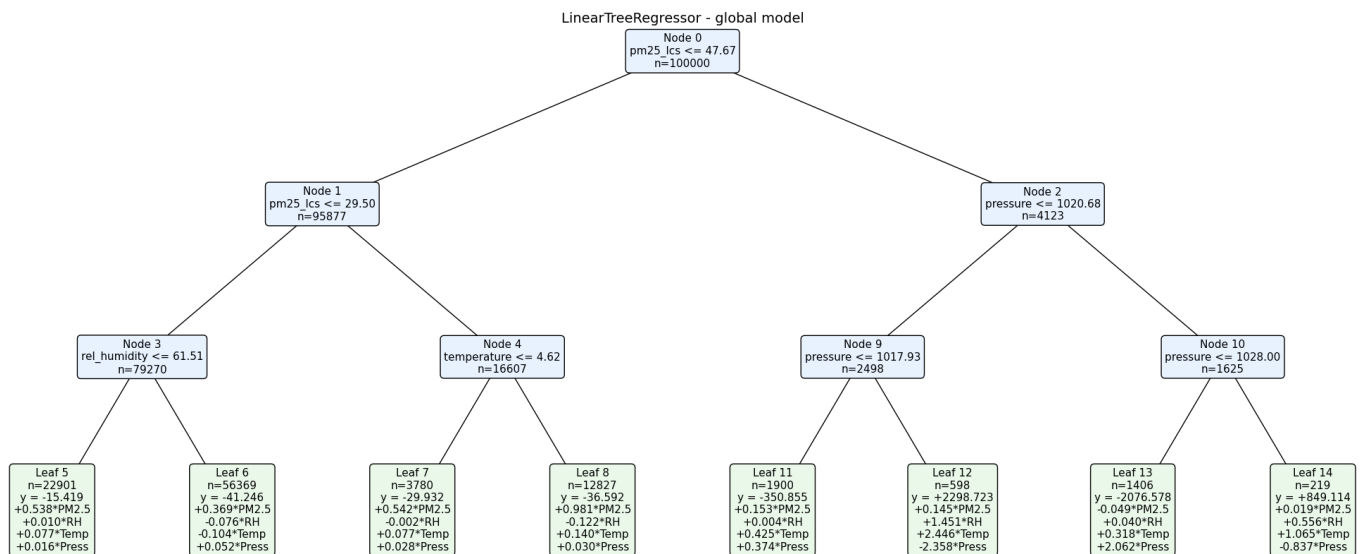


Figure 7. Global Linear Tree Regressor Visualization

The generated local linear models demonstrate the adaptive calibration behavior of the proposed machine learning framework. Example linear equations extracted from selected leaf nodes are shown below.

Note: In the leaf-level regression equations,  $PM_{2.5}$  refers to the low-cost sensor  $PM_{2.5}$  input variable ( $pm25\_lcs$ ). The equation uses this measurement together with relative humidity, temperature, and atmospheric pressure to estimate the calibrated  $PM_{2.5}$  concentration.

Leaf 5:

$$y = -15.419 + 0.538PM_{2.5} + 0.010RH + 0.077Temp + 0.016Press$$

Leaf 6:

$$y = -41.246 + 0.369PM_{2.5} - 0.076RH - 0.104Temp + 0.052Press$$

Leaf 12:

$$y = 2298.723 + 0.145PM_{2.5} + 1.451RH + 2.446Temp - 2.358Press$$

Where:

- $y$  – Predicted  $PM_{2.5}$  value ( $\mu g/m^3$ )
- $PM_{2.5}$  – Raw  $PM_{2.5}$  value ( $\mu g/m^3$ ) measured by the low-cost sensor
- $RH$  – Relative humidity measurement (%)
- $Temp$  – Temperature value ( $^{\circ}C$ )
- $Press$  – Atmospheric pressure value (hPa)

The complete set of decision rules and local linear regression equations generated by the final Linear Tree Regressor model is provided in Appendix C.

From these local regression equations, it can be observed that the relationship between  $PM_{2.5}$  measurements and environmental variables was different across various regions of the feature space. The modified Linear Tree Regressor thus dynamically adjusted calibration behavior based on local environmental conditions, thus allowing for improved prediction flexibility over the static EPA correction equations.

Another important observation was the stability of the machine learning model across cross-validation folds. The fold-to-fold standard deviations were relatively small, indicating consistent predictive performance across different train-test splits and sensor-reference station combinations.

Below *Figure 8* shows a time-series comparison between the EPA correction method, the Linear Tree Regressor model predictions, and the reference  $PM_{2.5}$  observations. It should be noted that this figure is intended as a qualitative visualization of model behavior rather than a formal performance evaluation. The Linear Tree Regressor predictions shown in the figure were generated using the final trained model applied to the full dataset and do not correspond to a separate cross-validation test fold. Quantitative performance assessment was conducted independently using stratified five-fold cross-validation, and the reported  $MAE$ ,  $RMSE$ , and  $R^2$  values are based on those cross-validation results.

The Linear Tree Regressor generally followed the temporal patterns of the reference  $PM_{2.5}$  observations more closely and produced fewer large deviations than the EPA correction baseline. In contrast, the EPA correction method exhibited larger fluctuations and several overestimation spikes during periods of elevated  $PM_{2.5}$  concentrations.

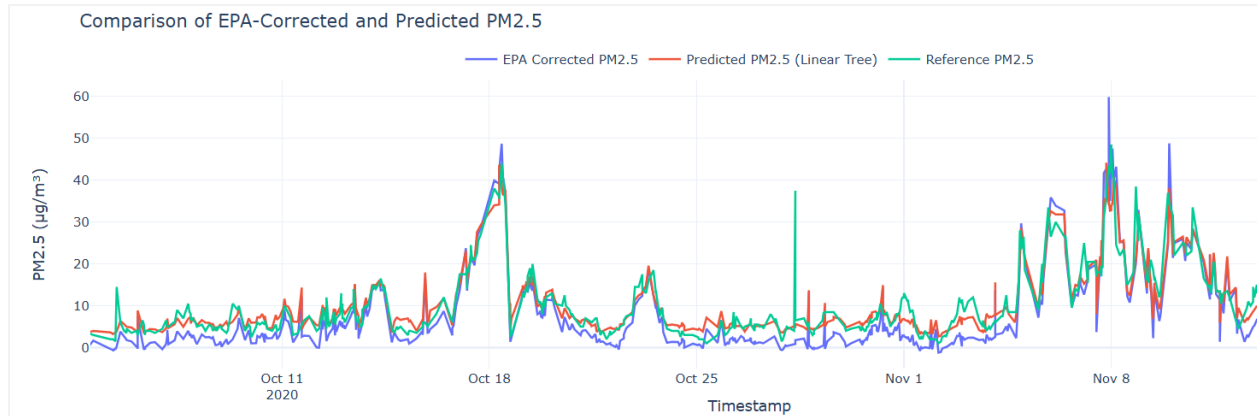


Figure 8. Illustrative time-series comparison between EPA-corrected  $PM_{2.5}$  values, predictions from the final Linear Tree Regressor model, and reference monitoring station observations. The figure is presented for visualization purposes only; quantitative performance evaluation was conducted using stratified five-fold cross-validation.

Overall, the results strongly support the conclusion that machine-learning-based post-hoc calibration can considerably improve the reliability of low-cost  $PM_{2.5}$  sensor measurements. These results further show that relatively interpretable machine learning models such as the Linear Tree Regressor can be calibrated effectively without the need of complex deep learning architectures.

### 4.3 RQ2 – Reference Station Optimization

The second experiment investigated how the number of reference monitoring stations affected calibration performance. The objective was to determine the minimum number of reference stations required to achieve reliable  $PM_{2.5}$  calibration while reducing dependence on expensive regulatory monitoring infrastructure.

#### 4.3.1 Subset Experiment Results

To evaluate the effect of reference station quantity, multiple subsets of reference monitoring stations were generated using subset sizes ranging from one to nine stations. For each subset configuration, a new Linear Tree Regressor model was trained using only sensor-reference station pairs associated with the selected reference stations and evaluated using a common testing dataset. Calibration performance was measured primarily using Mean Absolute Error (MAE).

The objective of this experiment was to assess the influence of the number of reference monitoring stations on calibration performance. Spatial distance was not used as a selection criterion when generating the subsets. Consequently, the subsets contained sensor-reference station pairs spanning a range of distances, reflecting the natural spatial distribution of the monitoring network. The effect of spatial distance was investigated separately in RQ3.

The results showed that calibration performance improved gradually as the number of reference monitoring stations increased. However, the magnitude of improvement remained relatively small across the different subset sizes. Using only one reference station already achieved a

relatively low *MAE* of approximately  $2.39 \mu\text{g}/\text{m}^3$ , while using all nine reference stations reduced the *MAE* only slightly to approximately  $2.31 \mu\text{g}/\text{m}^3$ .

For a given subset size ( $k$ ), the reference monitoring stations were selected using random subset generation rather than nearest-station selection. For example, when ( $k = 1$ ), a single reference station was randomly selected from the nine available stations. When ( $k > 1$ ), multiple random combinations of reference stations were generated and evaluated. The reported *MAE* values therefore represent the average calibration performance across the sampled subsets for each subset size rather than the performance of a specific reference station configuration.

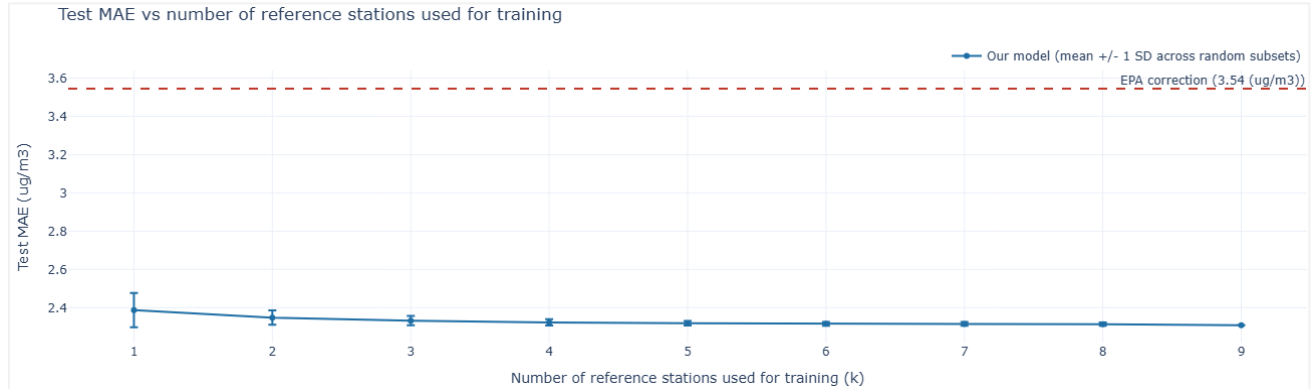


Figure 9. MAE vs Number of Reference Stations

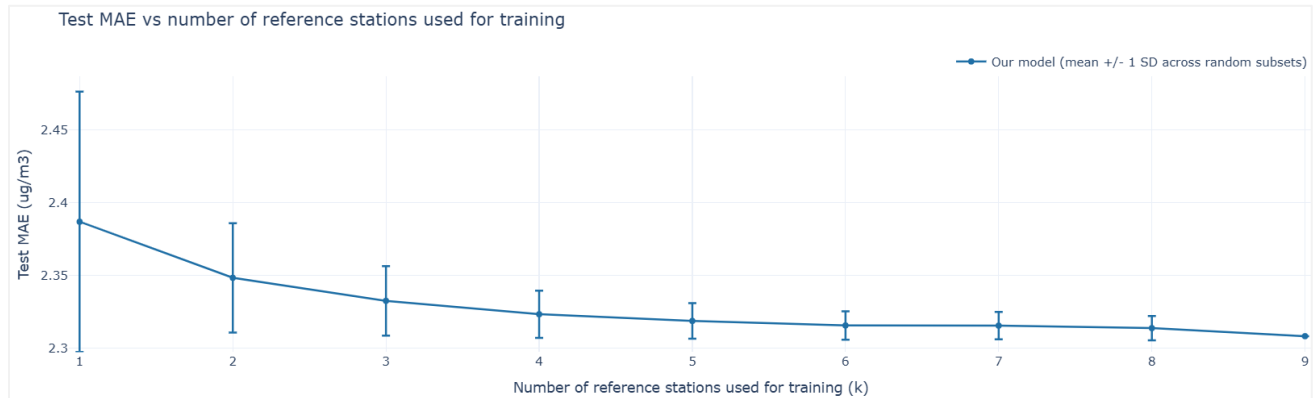


Figure 10. MAE vs Number of Reference Stations – Zoom In the Model

Table 7. Reference Station Subset Results

| <b>k</b> | <b>MAE Mean</b> | <b>MAE SD</b> | <b>Number of Reference Stations (n)</b> |
|----------|-----------------|---------------|-----------------------------------------|
| 1        | 2.387           | 0.089         | 9                                       |
| 2        | 2.348           | 0.038         | 36                                      |
| 3        | 2.332           | 0.024         | 84                                      |
| 4        | 2.323           | 0.016         | 126                                     |
| 5        | 2.319           | 0.012         | 126                                     |
| 6        | 2.315           | 0.010         | 84                                      |
| 7        | 2.315           | 0.009         | 36                                      |
| 8        | 2.314           | 0.008         | 9                                       |
| 9        | 2.308           | 0.000         | 1                                       |

EPA correction MAE on test set:  $3.54 \mu\text{g}/\text{m}^3$

The largest improvement was observed when increasing the number of stations from one to three, where the MAE decreased from approximately  $2.387 \mu\text{g}/\text{m}^3$  to  $2.332 \mu\text{g}/\text{m}^3$ . Beyond approximately five reference stations, the performance improvements became minimal, indicating a clear performance plateau (See *Figure 9*, *Figure 10*, and *Table 7*).

In comparison, the EPA correction baseline achieved a substantially larger MAE of approximately  $3.54 \mu\text{g}/\text{m}^3$  on the same testing dataset. Even the smallest subset configuration using a single reference station significantly outperformed the EPA correction method.

### 4.3.2 Interpretation

The findings indicate that acceptable calibration performance can be achieved even when using only a single reference monitoring station. This result is important because reference stations are expensive to install and maintain, particularly in large urban monitoring networks.

The observed performance plateau suggests that strategic placement of reference monitoring stations may be more important than simply increasing the total number of stations. Well-distributed reference stations may provide sufficient environmental representation for effective machine learning-based calibration while significantly reducing infrastructure costs.

Overall, the experiment demonstrates the potential scalability of the proposed calibration framework and highlights the feasibility of deploying cost-effective urban air quality monitoring systems using a limited number of regulatory-grade reference stations.

## 4.4 RQ3 – Distance Analysis

### 4.4.1 Distance-Based Error Trends

The third experiment investigated the relationship between spatial distance and calibration accuracy. To evaluate this, the average Mean Absolute Error (MAE) was analyzed across multiple distance intervals between low-cost sensors and reference monitoring stations.

*Figure 11* illustrates the mean prediction error trends across increasing spatial distance intervals for both the EPA correction method and the Linear Tree Regressor model.

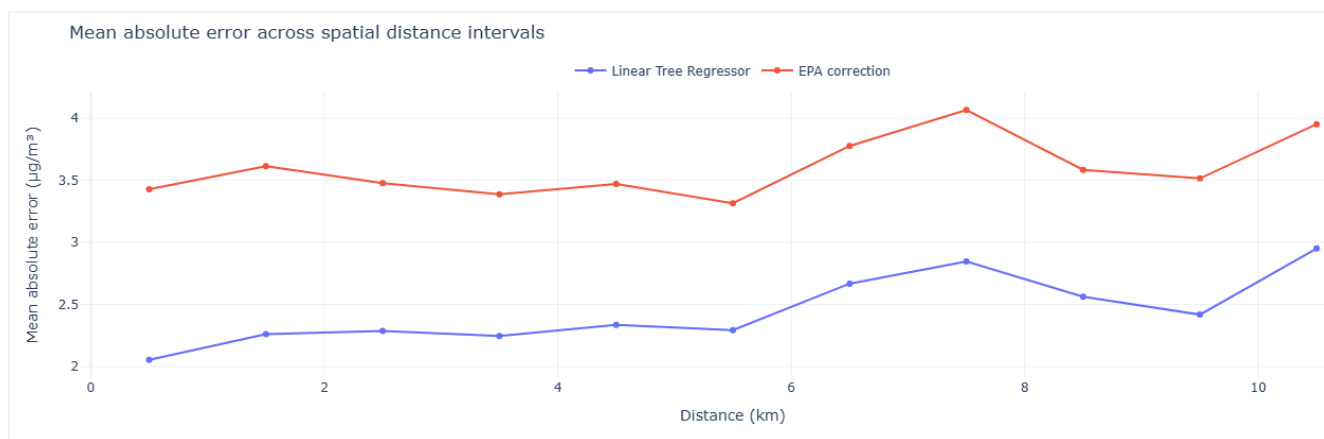


Figure 11. Mean absolute error across spatial distance intervals for the EPA correction method and the Linear Tree Regressor model

The results showed that the Linear Tree Regressor model consistently produced substantially lower prediction errors than the EPA correction method across all evaluated distance intervals. The machine learning model maintained *MAE* values primarily between approximately 2.0 and 3.0  $\mu\text{g}/\text{m}^3$ , whereas the EPA correction method generally exhibited larger errors ranging from approximately 3.3 to 4.1  $\mu\text{g}/\text{m}^3$ .

Although some variability in prediction error was observed across the spatial distance ranges, both calibration approaches demonstrated a gradual increase in prediction error as distance increased. The increase was more noticeable for the Linear Tree Regressor model, particularly at larger spatial separation distances.

Overall, the distance-based analysis demonstrated that the proposed machine learning framework maintained superior calibration accuracy across all evaluated spatial distance intervals despite exhibiting greater sensitivity to increasing sensor-reference station separation distances.

#### 4.4.2 Regression Analysis

Linear regression analysis was performed to evaluate how calibration error changed with increasing spatial distance between low-cost sensors and reference monitoring stations for both the EPA correction method and the Linear Tree Regressor model.

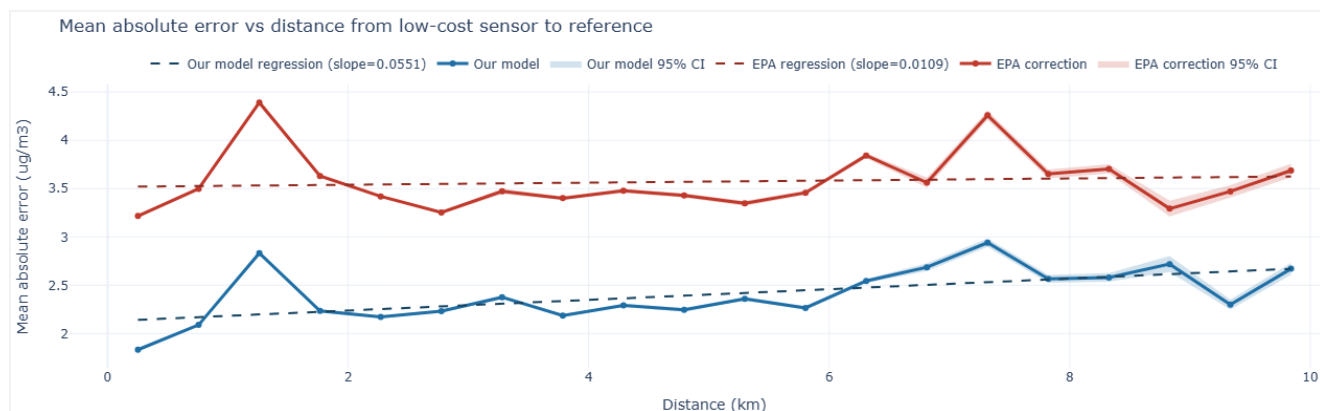


Figure 12. Relationship between spatial distance and mean absolute error for the EPA correction method and the Linear Tree Regressor model

*Figure 12* illustrates the distance-error regression trends for both calibration approaches. The EPA correction method exhibited a relatively small positive slope, indicating a weaker overall relationship between distance and prediction error. In contrast, the Linear Tree Regressor demonstrated a steeper positive average trend, suggesting that prediction error increased more noticeably with increasing spatial separation.

Although the overall regression slope is positive for both calibration methods, *Figure 12* shows substantial variability between distance bins. For example, the EPA correction method exhibits a pronounced error increase around 1–1.5 km that is not observed in neighboring bins. Such fluctuations are expected because each distance bin contains different sensor-reference station combinations and environmental conditions. Therefore, individual peaks should not be interpreted as evidence of a specific distance threshold. Instead, the regression lines represent the

overall average trend across all distance bins, indicating that calibration error tends to increase gradually with distance despite local variability.

The estimated regression slope for the EPA correction method was approximately  $0.02 \mu\text{g}/\text{m}^3$  per kilometer, whereas the Linear Tree Regressor exhibited a larger slope of approximately  $0.06 \mu\text{g}/\text{m}^3$  per kilometer. These findings suggest that although the proposed machine learning model achieved substantially lower overall prediction error, its calibration performance degraded more rapidly as the distance between low-cost sensors and reference monitoring stations increased.

Despite the stronger distance sensitivity, the Linear Tree Regressor maintained substantially lower prediction error than the EPA correction method across the entire observed distance range (approximately 0–10 km) in the dataset.

#### 4.4.3 ANCOVA Results

ANCOVA was conducted to statistically evaluate the effects of spatial distance, calibration method, and their interaction on prediction error (see *Table 8*). ANCOVA was selected as the primary statistical framework because it simultaneously evaluates the effects of distance, calibration method, and their interaction on prediction error within a single model.

ANCOVA was selected because it allows simultaneous evaluation of a continuous covariate (distance), a categorical factor (calibration method), and their interaction effect on prediction error.

Table 8. ANCOVA Summary Results

| Variable                | Coefficient                    | p-value   |
|-------------------------|--------------------------------|-----------|
| Distance                | $2.128 \times 10^{-5}$         | 0.477     |
| Model                   | $-1.35 \mu\text{g}/\text{m}^3$ | $< 0.001$ |
| Distance $\times$ Model | $4.274 \times 10^{-5}$         | 0.001     |

The analysis showed that the calibration method had a statistically significant effect on prediction error ( $p < 0.001$ ). The coefficient for the model variable indicates that, at zero spatial distance, the Linear Tree Regressor achieved approximately  $1.35 \mu\text{g}/\text{m}^3$  lower prediction error than the EPA correction method.

The main effect of distance was not statistically significant for the EPA correction method ( $p = 0.477$ ), suggesting that the EPA baseline exhibited a relatively weak relationship between distance and prediction error.

The interaction between distance and calibration method was statistically significant ( $p = 0.001$ ), indicating that the effect of distance differed between the two calibration approaches. Although the interaction coefficient ( $4.274 \times 10^{-5}$ ) is numerically small, the large sample size provides sufficient statistical power to detect relatively small effects. Therefore, statistical significance should not be interpreted as evidence of a large practical effect. Instead, the practical importance of the interaction is better understood through the distance-error slopes and trends shown in *Figure 12*.

The estimated error-distance slope for the EPA correction method was approximately  $0.021 \mu\text{g}/\text{m}^3$  per kilometer, whereas the Linear Tree Regressor exhibited a larger slope of

approximately  $0.064 \mu\text{g}/\text{m}^3$  per kilometer. The difference between these slopes was approximately  $0.043 \mu\text{g}/\text{m}^3$  per kilometer. This means that for every additional kilometer of separation between a low-cost sensor and a reference monitoring station, the prediction error of the Linear Tree Regressor increased about  $0.043 \mu\text{g}/\text{m}^3$  faster than the prediction error of the EPA correction method.

Using these slope estimates, the theoretical break-even distance between the two calibration approaches was estimated to be approximately 31.7 km. This value was obtained by dividing the initial error advantage of the Linear Tree Regressor at zero distance ( $1.35 \mu\text{g}/\text{m}^3$ ) by the difference between the distance-error slopes ( $0.043 \mu\text{g}/\text{m}^3$  per kilometer). In practical terms, the break-even distance represents the point at which the lower initial error of the Linear Tree Regressor would theoretically be offset by its greater distance sensitivity, resulting in similar prediction errors for both calibration methods.

It should be noted that this break-even distance is a theoretical extrapolation derived from the ANCOVA model rather than a directly observed value. Since the study data covered distances of approximately 0–10 km, the 31.7 km estimate should be interpreted as an analytical indication of the relative distance sensitivity of the two calibration approaches rather than as a validated deployment threshold.

Overall, the ANCOVA results demonstrate that the Linear Tree Regressor significantly improved calibration accuracy compared with the EPA correction method, while also exhibiting greater sensitivity to increasing spatial distance.

## 4.5 Discussion of Findings

The experimental results demonstrate that machine learning-based post-hoc calibration can substantially improve the accuracy and reliability of low-cost  $\text{PM}_{2.5}$  sensor measurements. Across all evaluation experiments, the Linear Tree Regressor model consistently outperformed the EPA correction baseline by achieving lower prediction errors and higher explanatory performance.

The reference station optimization experiments further indicated that effective calibration performance can be achieved using a relatively small number of reference monitoring stations. Even a single reference station produced substantially lower prediction error than the EPA correction baseline. The observed performance plateau suggests that strategic placement of reference stations may be more important than simply increasing the total number of stations within a monitoring network.

The spatial distance analysis showed that the relationship between distance and calibration error differed between the EPA correction method and the proposed machine learning framework. Although spatial distance alone was not statistically significant in the ANCOVA analysis, the interaction between calibration method and distance demonstrated that the Linear Tree Regressor achieved substantially lower overall calibration error than the EPA correction method, while its prediction error increased more rapidly with increasing spatial distance.

Despite this stronger distance sensitivity, the proposed machine learning framework remained more accurate than the EPA correction baseline across all evaluated spatial distance intervals and most practical deployment distances observed in the dataset. The estimated break-even distance

between the two calibration approaches was approximately 31.7 km, indicating that the proposed model maintained superior calibration performance across the majority of realistic urban deployment scenarios.

Finally, the availability of multi-city measurements from Antwerp, Oslo and Zagreb offers the possibility to develop generalized calibration models, which could be adjusted to different urban and environmental conditions in the future. Overall, the results suggest the potential for a scalable and affordable ML-based air quality monitoring system by using low-cost sensors.

## 4.6 Summary

In this section, we presented the experimental results and analysis to evaluate the proposed machine-learning-based post-hoc calibration framework for low-cost PM<sub>2.5</sub> sensors. The results showed that the calibration accuracy of the Linear Tree Regressor is significantly better than the baseline of EPA correction. In addition, the section discussed reference station optimization, spatial distance effects, sensor error behavior, and the potential for cross-city model generalization in the future. The overall results support the effectiveness and scalability of machine learning approaches for improving low-cost urban air quality monitoring systems.

## 5. Discussion

### 5.1 Interpretation of Results

The experimental results demonstrate that machine learning-based post-hoc calibration can substantially improve the performance of low-cost  $\text{PM}_{2.5}$  sensors compared to traditional empirical correction approaches. Across all experiments, the Linear Tree Regressor model consistently achieved lower prediction errors and higher explanatory performance than the EPA correction baseline. These findings indicate that machine learning models are capable of capturing complex nonlinear relationships between low-cost sensor measurements and environmental conditions more effectively than fixed calibration equations.

The improved performance of the proposed model is likely related to its ability to partition the feature space into multiple environmental regimes and apply localized linear regression models within each partition. Environmental variables such as relative humidity, temperature, and atmospheric pressure can strongly influence the behavior of optical particulate matter sensors [4],[6]. In particular, humidity-related particle growth can significantly alter light-scattering behavior, leading to measurement instability [4],[6]. The Linear Tree Regressor was able to adapt to these changing environmental conditions more effectively than the static EPA correction formula.

The results further showed that the proposed model produced fewer extreme prediction deviations and more stable temporal behavior during elevated  $\text{PM}_{2.5}$  events. This suggests that machine learning-based calibration frameworks may be particularly beneficial during periods of rapidly changing environmental conditions or high pollution concentrations, where traditional empirical corrections may become less reliable.

The spatial analysis also provided important insights into calibration robustness. Although the Linear Tree Regressor model achieved substantially lower overall calibration error, the statistical analysis demonstrated that its prediction error increased significantly with increasing spatial distance between low-cost sensors and reference monitoring stations. In contrast, the EPA correction baseline exhibited relatively stable error across distance ranges. This finding suggests that the proposed machine learning framework may depend more strongly on localized environmental relationships learned during training.

Nevertheless, despite this stronger distance sensitivity, the proposed model remained more accurate than the EPA correction method across most practical urban deployment distances observed in the dataset. The estimated break-even distance of approximately 31.7 km indicates that the proposed machine learning framework remains highly suitable for realistic urban monitoring scenarios where sensor-reference station distances are typically much smaller.

### 5.2 Practical Implications

The results of this study provide several important practical insights for urban air quality monitoring and environmental sensing systems. Traditional reference monitoring stations provide highly accurate measurements but are expensive to install, operate, and maintain. Consequently, many cities and regions lack sufficiently dense monitoring networks capable of capturing fine-scale spatial variability in air pollution [5],[10].

The proposed machine learning calibration framework demonstrates that low-cost PM<sub>2.5</sub> sensor networks can achieve substantially improved accuracy through post-hoc calibration using reference monitoring station data. This creates opportunities for scalable and cost-effective deployment of dense urban air quality monitoring systems capable of providing improved spatial coverage compared to traditional reference station networks alone [3],[5],[10],[23],[24].

The experiments show that good calibration results can be achieved with only a few carefully placed reference stations. This is useful for cities that do not have many environmental monitoring systems. It is also helpful for cities with limited budgets. Rather than maximizing the number of reference stations, calibration performance may depend more strongly on optimal spatial placement and representativeness of environmental conditions.

The observed spatial distance effects also provide important guidance for future deployment strategies. Since calibration accuracy decreased with increasing distance for the proposed machine learning framework, future monitoring networks should consider maintaining reasonable spatial proximity between low-cost sensors and reference stations whenever possible.

Overall, the proposed framework may support the development of more accessible and geographically distributed air quality monitoring systems capable of improving environmental awareness, pollution exposure assessment, and public health protection.

### 5.3 Contribution to Research

This thesis contributes to the expanding field of low-cost air quality sensor calibration and machine learning-based environmental monitoring in several significant ways.

First, the study utilized a very large real-world dataset consisting of approximately 1.99 million observations generated from pairwise combinations between 9 reference monitoring stations and 31 low-cost sensor systems. Compared to many previous studies that rely on short-term co-location experiments or relatively small datasets, the large-scale nature of the dataset improved the robustness and generalizability of the experimental analysis.

Second, the study investigated spatial calibration behavior using ANCOVA. Many previous calibration studies focus primarily on overall prediction accuracy while ignoring spatial effects. In contrast, this thesis explicitly evaluated how calibration performance changes as the spatial distance between low-cost sensors and reference monitoring stations increases.

Third, the thesis investigated reference station optimization and demonstrated that effective calibration performance can potentially be achieved using a limited number of strategically selected reference monitoring stations. This finding has practical relevance for scalable urban monitoring network design.

Fourth, the study applied a Linear Tree Regressor model for PM<sub>2.5</sub> calibration. The model combines decision-tree partitioning with localized linear regression models, enabling both nonlinear environmental modeling and interpretability. This approach provided improved calibration performance while still allowing the calibration logic to remain interpretable through explicit leaf-level regression equations.

Finally, the study combined machine learning calibration with spatial statistical analysis and environmental interpretation, providing a more comprehensive evaluation framework for low-cost air quality monitoring systems.

## 5.4 Limitations

Several limitations of this study should be considered when interpreting the results and overall findings.

First, the SenseURCity dataset contains deployments from Antwerp, Oslo and Zagreb, but the main experiments were conducted exclusively on the Antwerp deployment, and therefore the experimental findings may not generalize fully to regions with substantially different climatic, meteorological or urban conditions.

Second, seasonal variability may affect calibration behavior. Environmental conditions such as humidity, temperature and particulate composition, can vary significantly between seasons and may influence sensor performance differently at different times of the year [4],[6],[20],[21].

Third, the long-term sensor drift has not been explicitly studied in this work. Optical particulate matter sensors may suffer from gradual performance degradation over time due to aging, contamination or hardware instability [3],[5],[21]. Future work should investigate adaptive calibration strategies capable of compensating temporal sensor drift.

Fourth, although the spatial analysis examined the relationship between calibration performance and sensor-reference station distance, other spatial variables such as traffic density, land-use characteristics, wind patterns, and local emission sources were not explicitly included in the analysis [8],[9],[23]. Furthermore, the ANCOVA results should be interpreted with caution because the large sample size may lead to statistically significant results for relatively small effect sizes, and unmeasured environmental factors may also influence calibration error.

The ANCOVA model used in the spatial analysis should also be interpreted with caution. Diagnostic statistics indicated departures from normality in the residual distribution and a relatively large condition number, suggesting potential numerical instability or correlation between explanatory variables. However, the primary objective of the ANCOVA analysis was explanatory rather than predictive. To improve the robustness of statistical inference, cluster-robust standard errors were applied using the sensor-reference station pair identifier. Consequently, the ANCOVA results should be interpreted as evidence of overall trends and associations rather than precise predictive relationships.

Finally, the proposed model was evaluated using supervised post-hoc calibration with available reference station measurements. In real-world deployments, reference station availability may be limited, and calibration transferability across cities or environmental contexts may require additional investigation [12],[22],[24].

## 5.5 Ethical and Environmental Considerations

The improved air quality monitoring discussed here has important ethical, environmental, and public health implications. More accurate low-cost sensor calibration can facilitate public health protection, urban sustainability initiatives, environmental policy development, and increased community awareness of air pollution exposure.

In particular, scalable low-cost monitoring networks may help underserved or under-monitored communities gain better access to localized air quality information. Enhanced spatial coverage can contribute to more informed environmental decision-making and support efforts to reduce health risks associated with long-term PM<sub>2.5</sub> exposure [3],[5].

In addition, dense low-cost sensor networks are also likely to improve transparency in environmental monitoring, and help policymakers to identify pollution hotspots, assess mitigation strategies, and support evidence-based decisions on urban planning [23],[24].

## 6. Conclusion

### 6.1 Summary of the Study

This thesis investigated a machine learning-based post-hoc calibration framework for improving the accuracy of low-cost  $\text{PM}_{2.5}$  sensors using regulatory-grade reference monitoring station data. A Linear Tree Regressor model was developed and evaluated using the SenseURCity dataset, with the primary experiments conducted on the Antwerp deployment.

The study addressed three main research questions related to calibration accuracy, reference station optimization, spatial distance effects, and model generalization potential. The proposed machine learning framework was compared against the EPA correction baseline using multiple statistical evaluation metrics and spatial analysis techniques.

### 6.2 Main Findings

The principal findings of this thesis can be summarized as follows:

1. The Linear Tree Regressor model significantly outperformed the U.S. Environmental EPA correction method by achieving lower *MAE* and *RMSE* values together with higher  $R^2$  performance across the cross-validation experiments.
2. Effective  $\text{PM}_{2.5}$  calibration performance could be achieved using a relatively small number of reference monitoring stations, indicating that strategic placement of stations may be more important than increasing the total number of stations within a monitoring network.
3. The spatial analysis demonstrated that the relationship between calibration error and distance differed between the EPA correction method and the Linear Tree Regressor model. Although the machine learning framework achieved substantially lower overall prediction error, its calibration error increased more rapidly with increasing spatial distance between low-cost sensors and reference monitoring stations.
4. The EPA correction method exhibited relatively stable prediction performance across increasing spatial distances, whereas the Linear Tree Regressor model demonstrated statistically significant distance sensitivity.
5. Despite the stronger spatial distance sensitivity, the proposed machine learning framework remained more accurate than the EPA correction baseline across most practical deployment distances observed in the dataset. The estimated break-even distance between the two calibration approaches was approximately 31.7 km.
6. The availability of multi-city measurements from Antwerp, Oslo, and Zagreb provides opportunities for future development of generalized and transferable machine learning-based  $\text{PM}_{2.5}$  calibration frameworks.

### 6.3 Final Conclusion

Results of this thesis show that machine learning methods can lead to significant improvement of calibration accuracy and reliability of low-cost  $\text{PM}_{2.5}$  sensors as compared to traditional

empirical correction methods. The Linear Tree Regressor framework was able to model complex environmental interactions and consistently improved calibration performance across several evaluation experiments.

The spatial analysis further demonstrated that although the proposed machine learning framework achieved superior overall calibration accuracy, its prediction error increased more rapidly with increasing sensor-reference station distance than the EPA correction baseline. Nevertheless, the proposed model remained more accurate across most practical urban deployment distances observed in this study, with an estimated break-even distance of approximately 31.7 km.

Overall, the study highlights the potential of machine learning-based post-hoc calibration as a cost-effective and scalable solution for urban air quality monitoring, especially in cities with limited regulatory monitoring infrastructure. The findings also demonstrate the importance of considering both calibration accuracy and spatial deployment characteristics when designing large-scale low-cost air quality monitoring networks.

## Future Work

Future research can further extend the proposed machine learning-based calibration framework in several important directions. One possible direction is cross-city model transferability, where calibration models trained using the Antwerp deployment are evaluated on the Oslo and Zagreb deployments. Such experiments would provide a deeper understanding of model generalization across different climatic, meteorological, and urban environments.

Another important direction is the development of generalized calibration models trained using data from multiple cities simultaneously. Comparing city-specific and generalized models would help determine whether a single calibration framework can be effectively applied across different deployment locations.

Future studies should also investigate long-term sensor drift and temporal model stability. The present study evaluated calibration performance within a common deployment period, but low-cost sensor performance may change over time because of sensor aging, contamination, and hardware degradation. Adaptive recalibration strategies and periodic model updating could therefore be explored.

Finally, future work could investigate the inclusion of additional spatial information within the calibration framework. Although the present study examined sensor-reference station distance as a spatial factor, future research may evaluate whether explicitly incorporating distance or other spatial variables as model inputs improves calibration performance and spatial robustness.

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# Appendices

## Appendix A – EPA Correction Equations

| Correction algorithm from the US EPA                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Developed by the US EPA for PM <sub>2.5</sub> from PMS5003 sensors | <p><b>AG<sub>raw</sub> &lt;30:</b></p> $PM_{2.5} = [0.524 \times AG_{raw}] - [0.0862 \times RH_{raw}] + 5.75$ <p><b>30 ≤ AG<sub>raw</sub> &lt;50:</b></p> $PM_{2.5} = [0.786 \times (AG_{raw}/20 - 3/2) + 0.524 \times (1 - (AG_{raw}/20 - 3/2))] \times AG_{raw} - [0.0862 \times RH_{raw}] + 5.75$ <p><b>50 ≤ AG<sub>raw</sub> &lt;210:</b></p> $PM_{2.5} = [0.786 \times AG_{raw}] - [0.0862 \times RH_{raw}] + 5.75$ <p><b>210 ≤ AG<sub>raw</sub> &lt;260:</b></p> $PM_{2.5} = [0.69 \times (AG_{raw}/50 - 21/5) + 0.786 \times (1 - (AG_{raw}/50 - 21/5))] \times AG_{raw} - [0.0862 \times RH_{raw} \times (1 - (AG_{raw}/50 - 21/5))] + [2.966 \times (AG_{raw}/50 - 21/5)] + [5.75 \times (1 - (AG_{raw}/50 - 21/5))] + [8.84 \times (10^{-4}) \times AG_{raw}^2 \times (AG_{raw}/50 - 21/5)]$ <p><b>260 ≤ AG<sub>raw</sub>:</b></p> $PM_{2.5} = 2.966 + [0.69 \times AG_{raw}] + [8.84 \times 10^{-4} \times AG_{raw}^2]$ |

**Figure A1.** Original EPA PM<sub>2.5</sub> correction algorithm for Plantower PMS5003 sensors  
Source: Adapted from Barkjohn et al. (2021) [11].

## Appendix B – Machine Learning Hyperparameters

| Hyperparameter          | Value                                                           | Description                                                         |
|-------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| Model                   | LinearTreeRegressor                                             | Proposed machine learning calibration model                         |
| Base Estimator          | Linear Regression                                               | Linear regression model used within leaf nodes                      |
| max_depth               | 3                                                               | Maximum depth of the decision tree                                  |
| min_samples_split       | 5000                                                            | Minimum number of samples required to split a node                  |
| min_samples_leaf        | 80                                                              | Minimum number of samples required in a leaf node                   |
| max_bins                | 32                                                              | Maximum number of bins used for split candidate generation          |
| criterion               | mse                                                             | Mean Squared Error split criterion                                  |
| random_state            | 42                                                              | Random seed used for reproducibility                                |
| Cross-Validation        | stratified five-fold cross-validation                           | Group-based stratified five-fold cross-validation                   |
| Training Subsample Size | 100,000                                                         | Number of observations used for global tree visualization           |
| Features                | PM PM <sub>2.5</sub> , Relative Humidity, Temperature, Pressure | Input variables used for calibration                                |
| Target Variable         | Reference PM <sub>2.5</sub>                                     | Ground-truth PM <sub>2.5</sub> measurements from reference stations |

**Table B1.** Hyperparameters and configuration settings used for the Linear Tree Regressor calibration model.

## Appendix C – Complete Linear Tree Regressor Decision Rules and Leaf Equations

The final Linear Tree Regressor model partitions the feature space according to PM<sub>2.5</sub> concentration, relative humidity, temperature, and atmospheric pressure. Each terminal node contains a separate local linear regression equation used to estimate calibrated PM<sub>2.5</sub> concentrations.

The complete decision rules and corresponding local linear regression equations are presented below.

### Rule 1

If:  $PM_{2.5} \leq 29.50$  and  $RH \leq 61.51$

$$y = -15.419 + 0.538 PM_{2.5} + 0.010 RH + 0.077 Temp + 0.016 Press$$

### Rule 2

If:  $PM_{2.5} \leq 29.50$  and  $RH > 61.51$

$$y = -41.246 + 0.369 PM_{2.5} - 0.076 RH - 0.104 Temp + 0.052 Press$$

### Rule 3

If:  $29.50 < PM_{2.5} \leq 47.67$  and  $Temp \leq 4.62$

$$y = -29.932 + 0.542 PM_{2.5} - 0.002 RH + 0.077 Temp + 0.028 Press$$

### Rule 4

If:  $29.50 < PM_{2.5} \leq 47.67$  and  $Temp > 4.62$

$$y = -36.592 + 0.981 PM_{2.5} - 0.122 RH + 0.140 Temp + 0.030 Press$$

### Rule 5

If:  $PM_{2.5} > 47.67$  and  $Press \leq 1017.93$

$$y = -350.855 + 0.153 PM_{2.5} + 0.004 RH + 0.425 Temp + 0.374 Press$$

### Rule 6

If:  $PM_{2.5} > 47.67$  and  $1017.93 < Press \leq 1028.00$

$$y = 2298.723 + 0.145 PM_{2.5} + 1.451 RH + 2.446 Temp - 2.358 Press$$

### Rule 7

If:  $PM_{2.5} > 47.67$  and  $Press > 1028.00$

$$y = -2076.578 - 0.049 PM_{2.5} + 0.040 RH + 0.318 Temp + 2.062 Press$$

### Rule 8

If:  $PM_{2.5} > 47.67$  and  $Press > 1028.00$

$$y = 849.114 + 0.019 PM_{2.5} + 0.556 RH + 1.065 Temp - 0.837 Press$$

Where:

- $y$  – Predicted PM<sub>2.5</sub> (µg/m<sup>3</sup>) value
- $PM_{2.5}$  – Raw PM<sub>2.5</sub> (µg/m<sup>3</sup>) value by the low-cost sensor
- $RH$  – Relative humidity measurement (%)
- $Temp$  – Temperature (°C) value
- $Press$  – Atmospheric pressure (hPa) value

## Appendix D – GitHub Repository

The complete implementation code and supplementary materials for this thesis are available at:

<https://github.com/udayakkarunarathna/pm25-sensor-calibration-framework> ,  
<https://github.com/AnjanaLakshan/pm25-sensor-calibration-framework>